

Week — Spread & distributions

Business Forecasting

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Where we are

C1 Framing
Forecasting → **C2 Describing data** → C3 Uncertainty → C4–C5 Regression → C6

We are wrapping up **C2 — describing data**: this week we cover both the **spread** of the data and the **shape** of its distribution.

- **Spread**: how dispersed is the data — **range, IQR, variance, standard deviation**, all summarized in a **boxplot**.
- **Distributions**: how the data is spread out — **histograms** and distribution **shape** (modality, skewness).

Population, parameters & statistics

Parameters vs Statistics

- You need to know how many people in Mexico have diabetes

Parameter

- Call μ_d the proportion of Mexican population which has diabetes
 - Usually the parameter is an **unknown** number **describing the whole population**
 - You want to learn what it is
 - In our example, μ_d is a parameter that you want to learn
 - More generally, parameter describes an aspect of the entire population

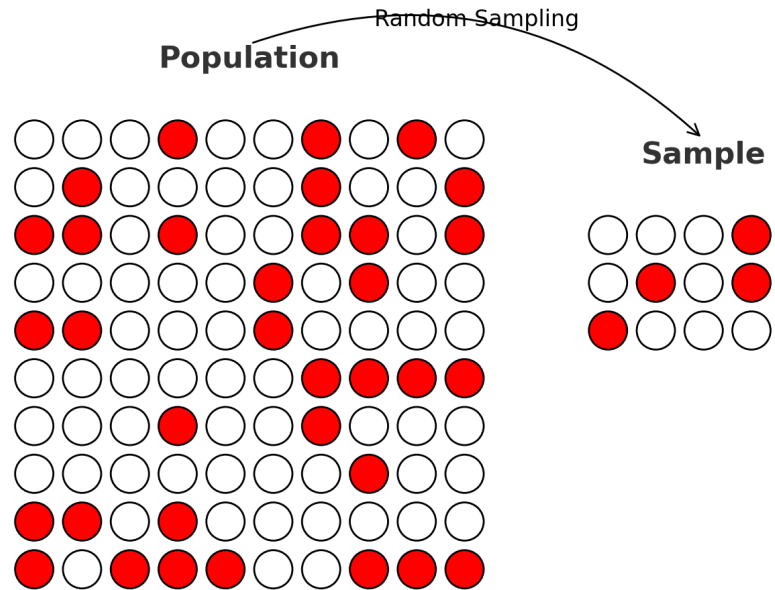
Parameters vs Statistics (cont.)

- You need to know how many people in Mexico have diabetes

Statistic

- But you don't have data on the whole population. At best you can get a sample from a survey
 - So you will try to estimate this parameter with sample
 - Statistic is a **guess of the parameter** which can be **calculated from the sample**
 - You will calculate a statistic $\hat{\mu}_p$ which is the proportion of diabetics in the sample

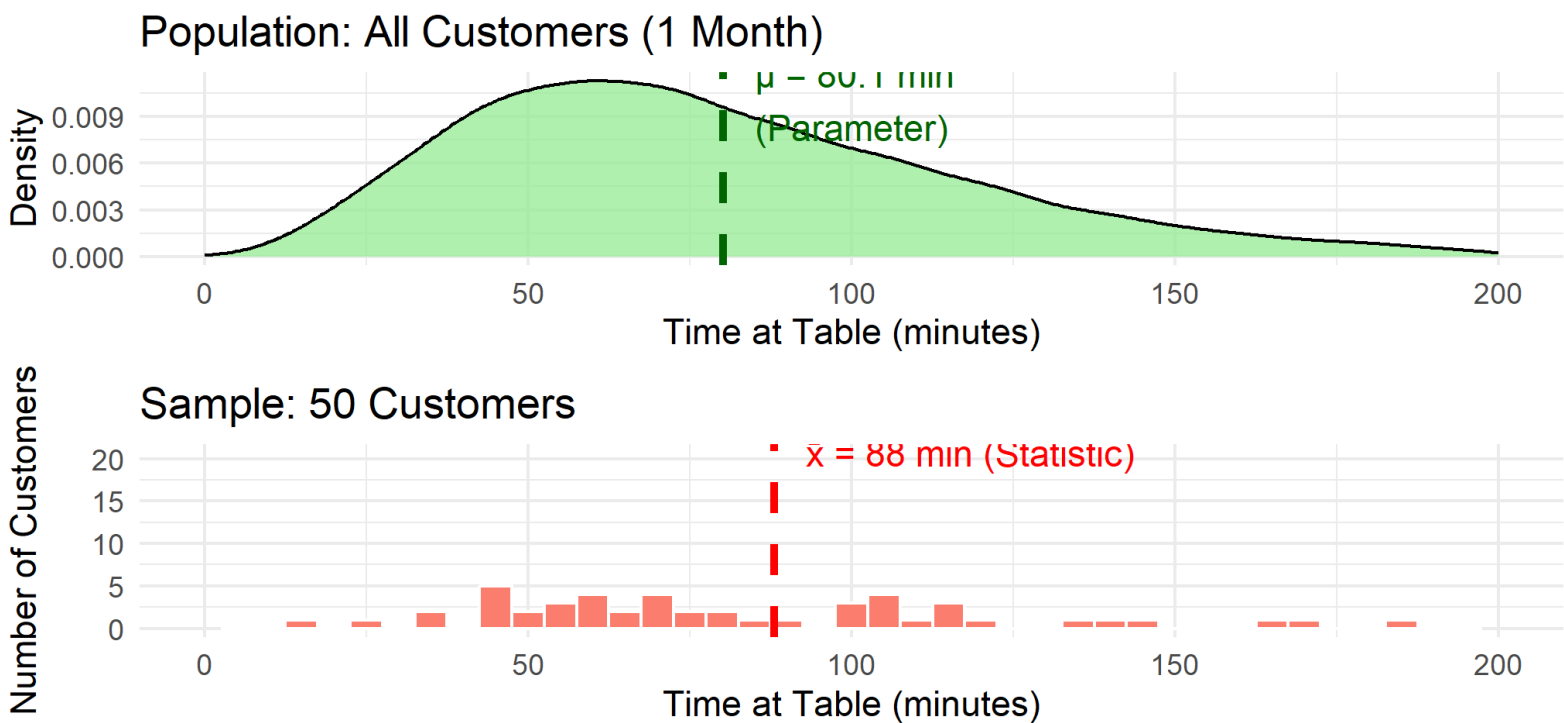
Sampling: categorical variable



Parameter (Population Diabetes %): 34.00%Statistic (Sample Diabetes %): 33.33%

Sampling: continuous variable

Customer Time at Table: Population vs. Sample Mean



Parameters vs Statistics

- What is population, sample, parameter and statistic in the following examples?
- You want to know the probability that a user who got a match on tinder will go out on a date with that person. You survey 1000 users and ask them about each match they got if they went on a date. You then calculate the share of dates which ended up in a match for these users.
- You want to know what whether starbucks baristas are faster than Cielito Querido baristas. You go to 10 starbucks and 10 Cielito Querido and measure the time it takes to make a coffee. You then calculate the average time it takes to make a coffee in each of these chains.
- You want to know the average age of people who go to the gym. You go to a gym and ask 100 people about their age. You then calculate the average age of these people.
- You want to know the variance of internet speed during in Mexico City. You visit 500 households and calculate the variance of their internet speed.

Parameters vs Statistics

- What are some examples of the statistics we will use and why do we care?

Measures of Dispersion

- Suppose a store has an average daily revenue of 10 000 pesos
- It could be that on each day it has exactly 10 000 pesos revenue
- Or it could be that on half of days it gets 20 000 pesos but on other half it gets 0 pesos
- Dispersion makes a big difference, especially when trying to understand risk!

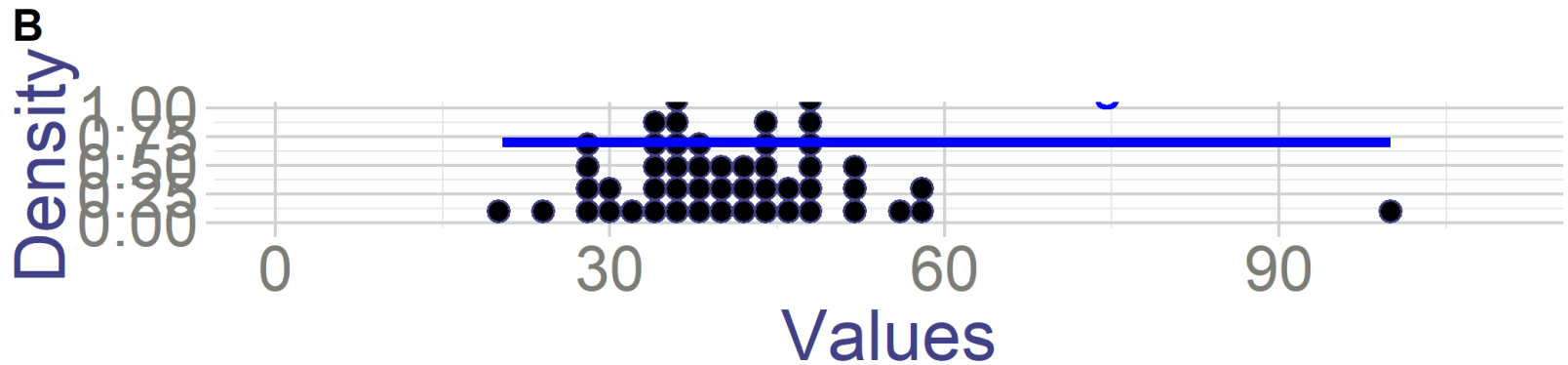
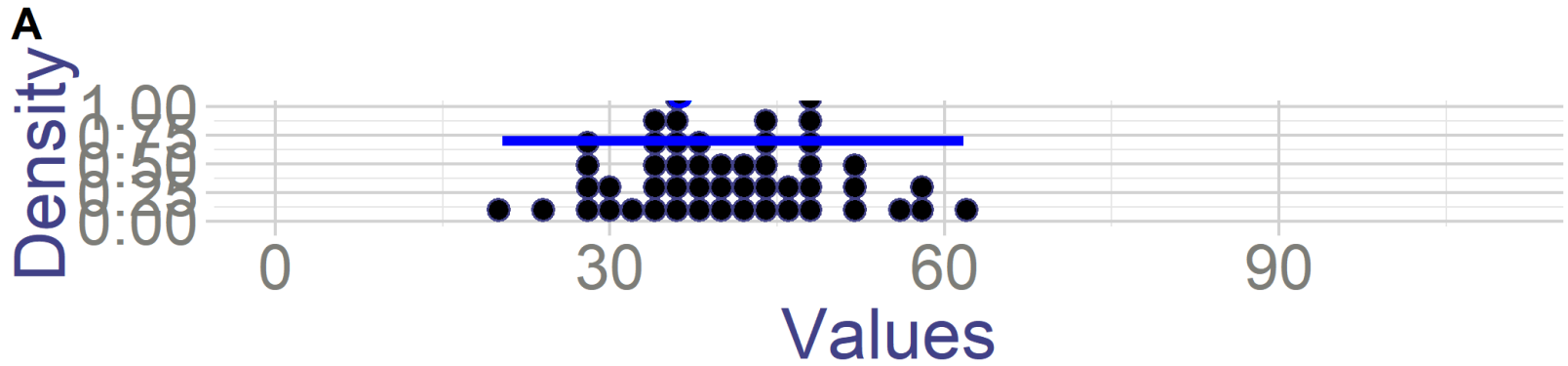
Range

- **Range** the difference between minimum and maximum value in the data

$$R = x_{max} - x_{min}$$

- What is the difference between the oldest and the youngest person with diabetes?
- **R**=77=97-20

- Very sensitive to outliers



Interquartile Range

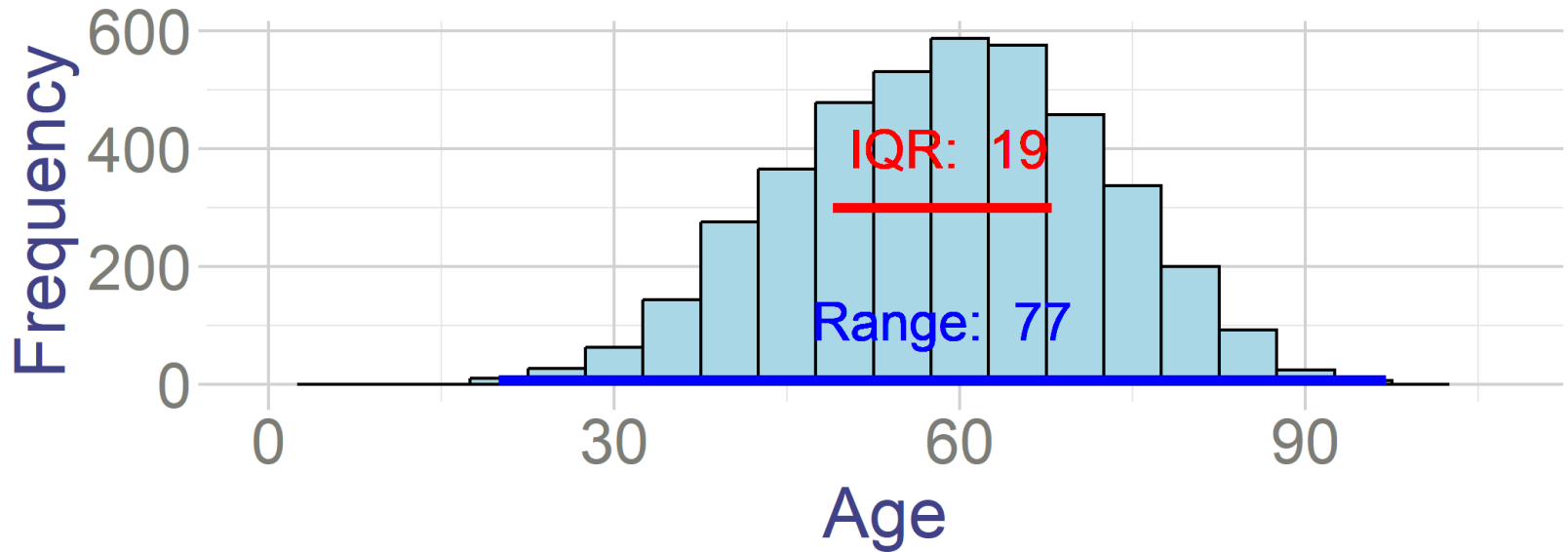
- **Interquartile range** is the difference between the first and the third quartile of the data:

$$IQR = q_3 - q_1$$

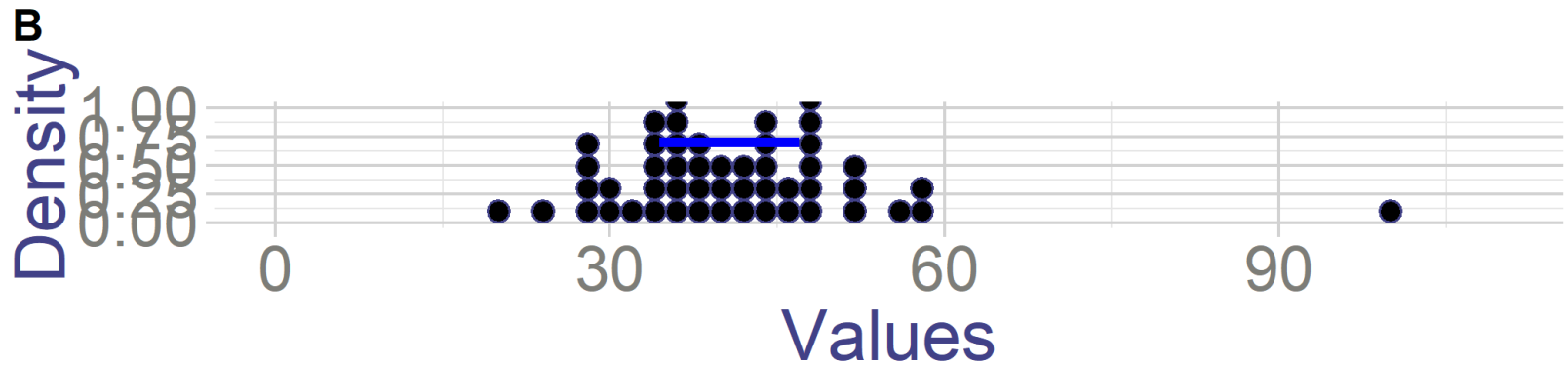
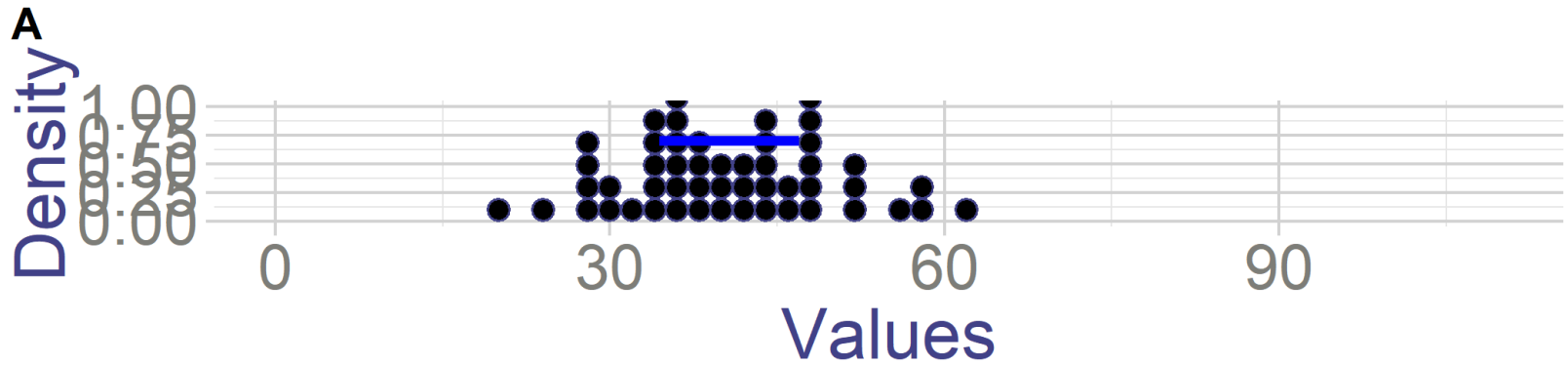
- What is the IQR of age in people with diabetes?
- **IQR**=19=68-49
- 50% of the sample is between q_3 and q_1

Interquartile Range (cont.)

- 50% of the sample (age of people with diabetes) is between q_1 and q_3



- Is it more or less sensitive to outliers than range?



Example with data

- What is the IQR?

Show

4

 entries

VideoTitle	Views
TikTok Video 1	30
TikTok Video 2	17
TikTok Video 3	22
TikTok Video 4	24

Showing 1 to 4 of 20 entries

Example with data

Here is a (smaller) data on distribution of how many views have various tik-tok videos.

- Suppose that all views triples and 1000 additional people viewed them as well

$$y_i = 3x_i + 1000$$

- What is new IQR?

Show 4 entries

VideoTitle	OldViews	NewViews
TikTok Video 1	30	1090
TikTok Video 2	17	1051
TikTok Video 3	22	1066
TikTok Video 4	24	1072

Showing 1 to 4 of 20 entries

IQR

- Order of observations was not affected, so same observations correspond to the first and the third quartile

$$q_1^{New} = 3q_1^{Old} + 1000$$

$$q_3^{New} = 3q_3^{Old} + 1000$$

- And more generally, for

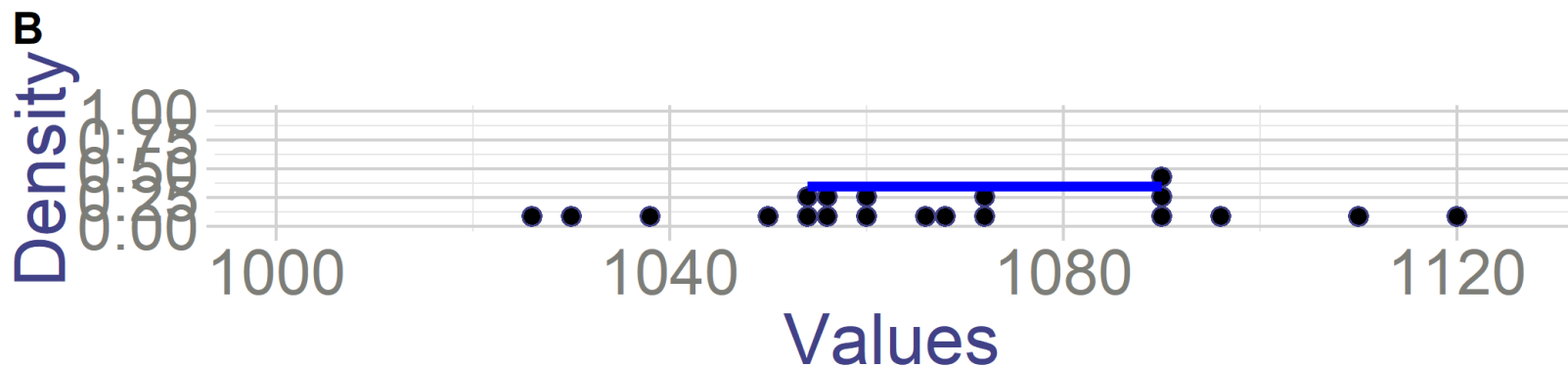
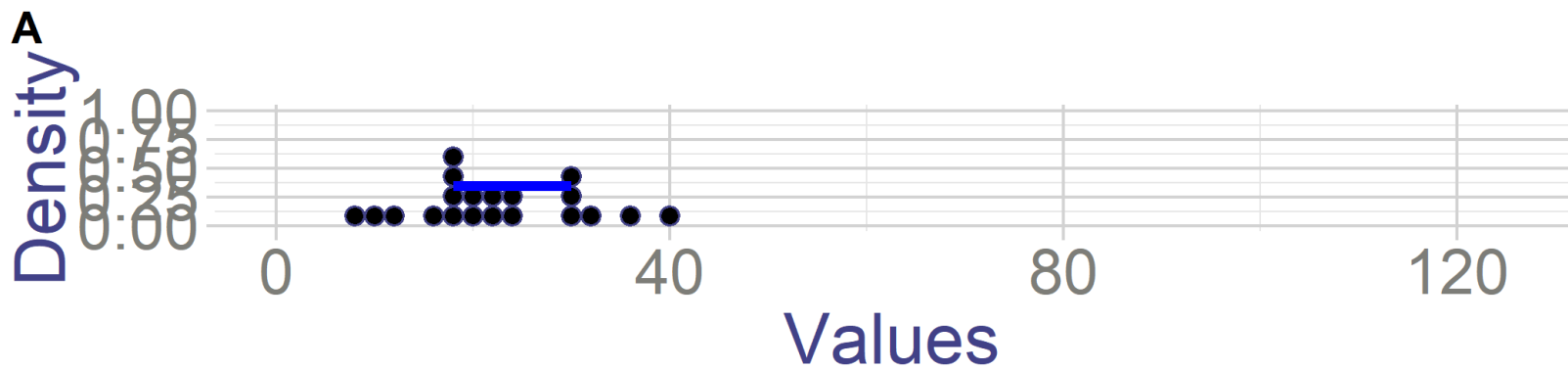
$$y_i = bx_i + a$$

and $b > 0$

$$v_p^y = bv_p^x + a$$

- if $b < 0$ then the order reverses.
- So what does it mean for IQR?

$$IQR^{New} = q_3^{New} - q_1^{New} = 3q_3^{Old} - 3q_1^{Old} = 3 * IQR^{Old}$$



Variance & Standard Deviation

Variance measures how far an average observation is from the mean:

- **Population variance:**

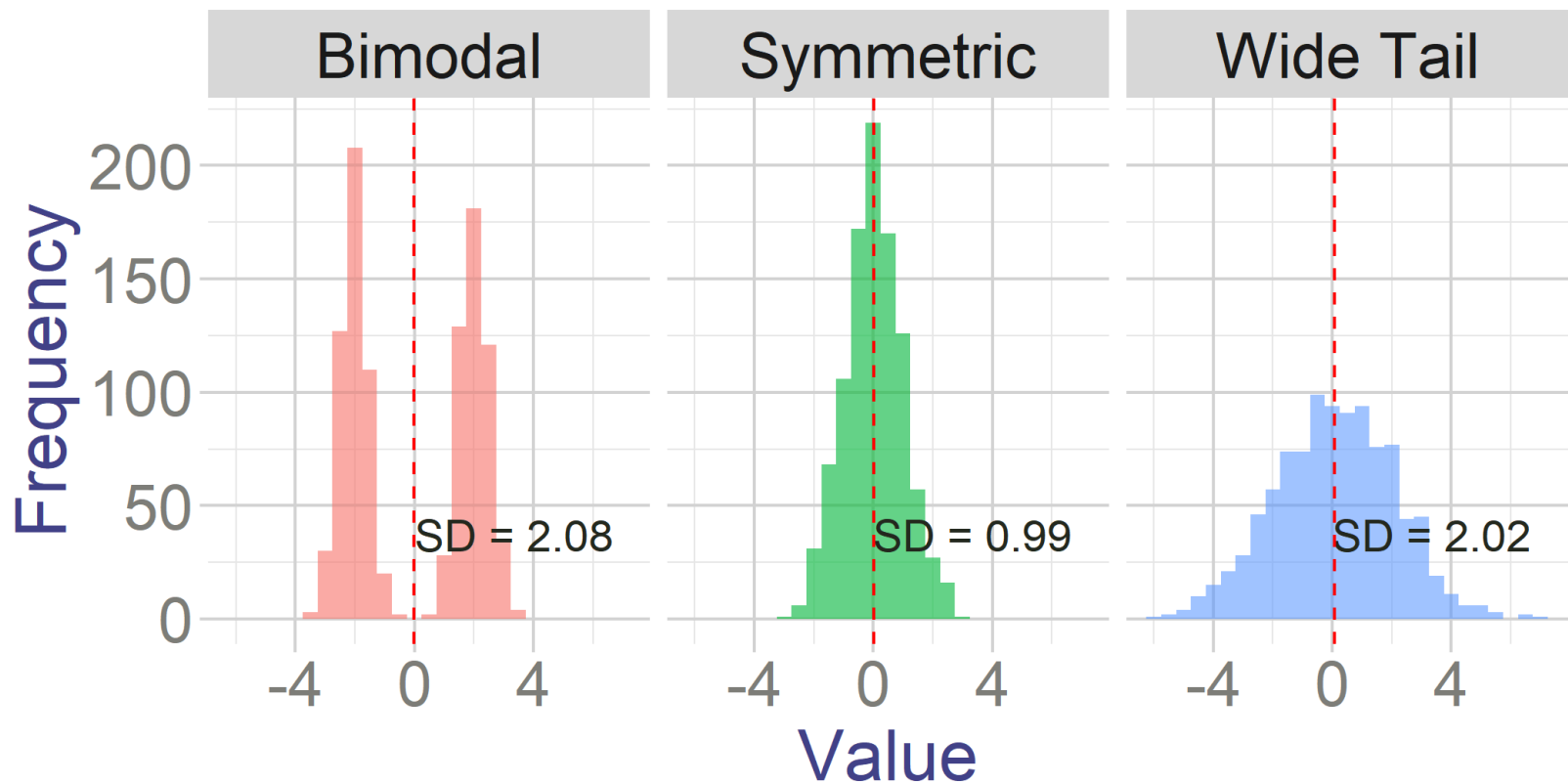
$$\sigma^2 = E[(X - \mu)^2] = E[X^2] - \mu^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2 = \frac{1}{N} \left(\sum_{i=1}^N x_i^2 - N\mu^2 \right)$$

For Discrete Variables it can be: $\sigma^2 = \sum_k P(X = k)(k - \mu)^2$, where k is any possible value that X can take.

- Variance does not have the right units since it squares everything, so we take the square root:
- **Population standard deviation** deviation:

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$$

Variance: same center, different spread



Variance & Standard Deviation

Consider two bets/situations:

- Bet A: with 75% you get 200 pesos and with 25% you lose me 200 pesos
 - 75% chance your life goes normal and you keep making money
 - 25% change your house burns down
- Bet B: with 75% you get 110 pesos and with 25% you get 70
 - When your life goes normal you get 110 (you pay 90 for insurance)
 - When your house burns down you are paid some compensation (70)
- Compute expected value and variance of each bet — which one would you prefer?
- What if I change Bet B to: with 75% you get 109 and with 25% you get 69? That's how insurance companies make profits.

Sample equivalents

- **Sample variance:**

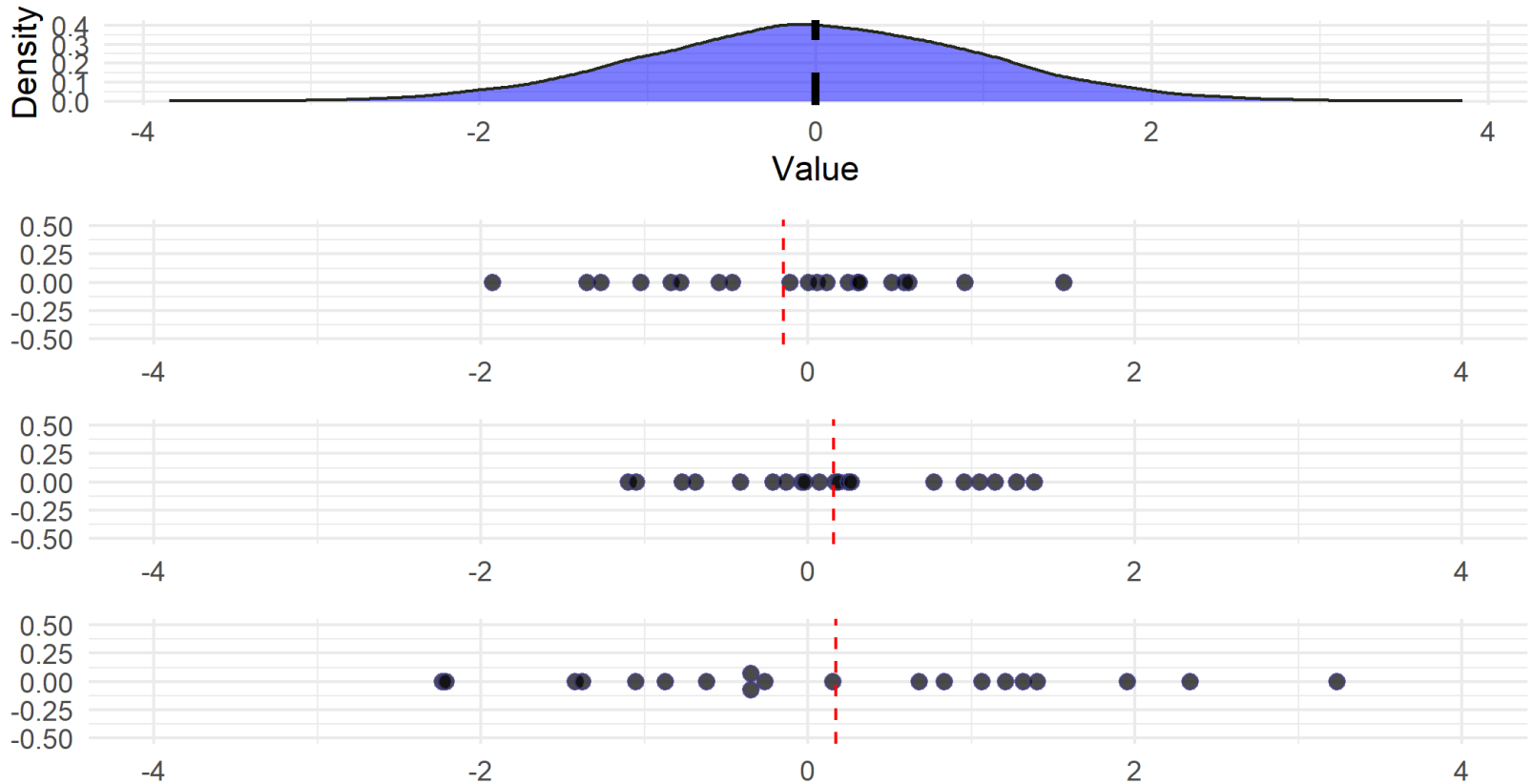
$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{1}{n-1} \left(\sum_{i=1}^n x_i^2 - n\bar{x}^2 \right)$$

- **Sample standard deviation** deviation:

$$s = \sqrt{\frac{1}{n-1} \left(\sum_{i=1}^n x_i^2 - n\bar{x}^2 \right)}$$

- Why we divide by $n - 1$ rather than n ?

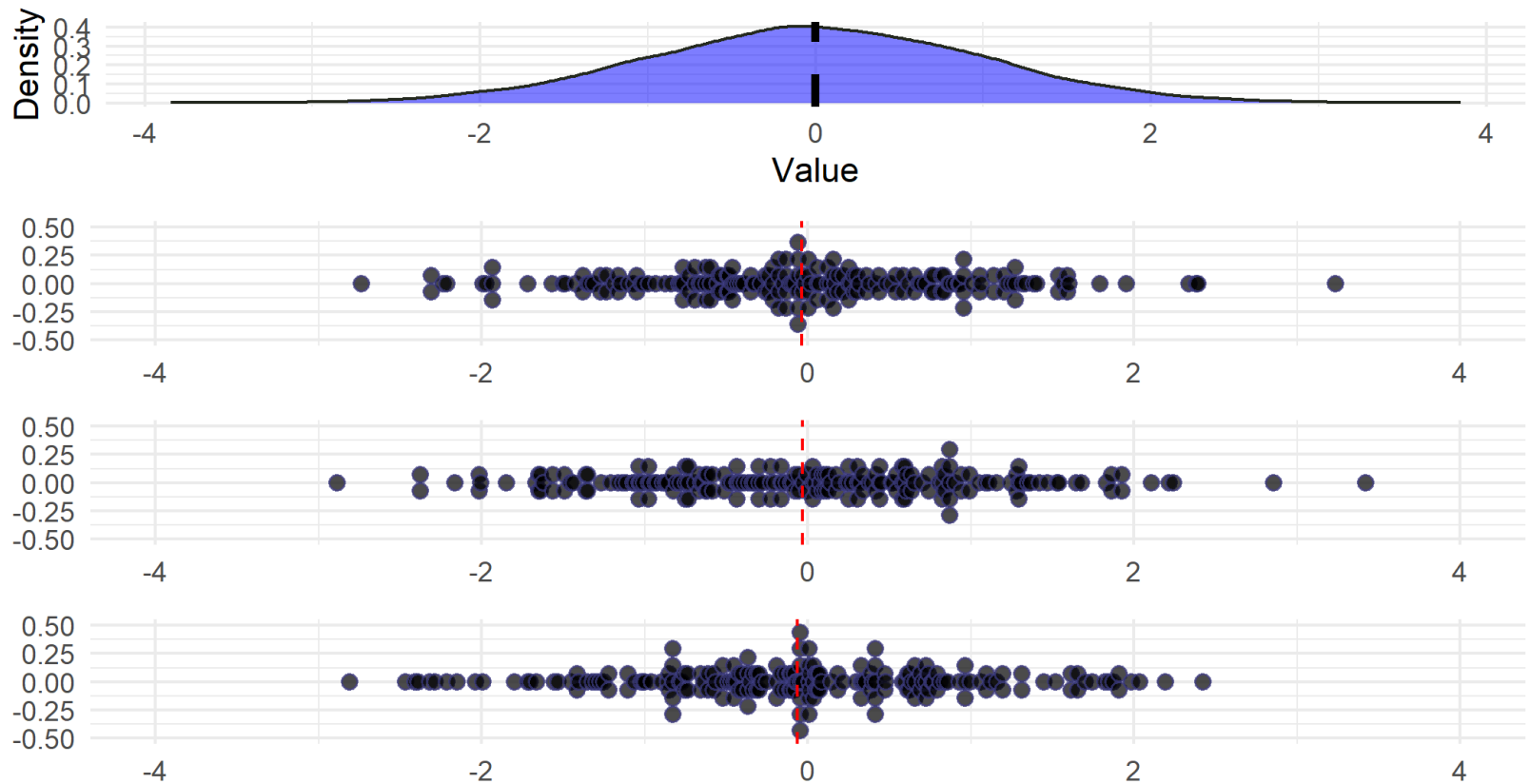
- **Intuition** - observed values usually fall closer to the sample mean than to the population mean. Distances are artificially small.



Sample equivalents

- So the deviations from the sample mean underestimate the population standard deviation
- So we divide by a smaller number to correct for it
- In big sample $\frac{1}{n}$ and $\frac{1}{n-1}$ are similar, so correction doesn't matter as much

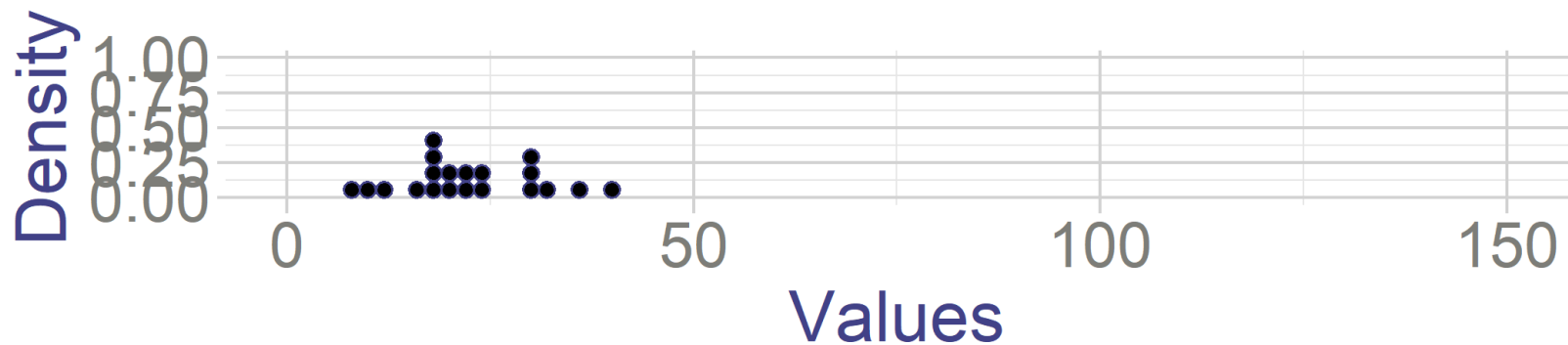
- **Intuition** - in big samples, our estimate of the population mean is already good, no need to correct



Standard Deviation

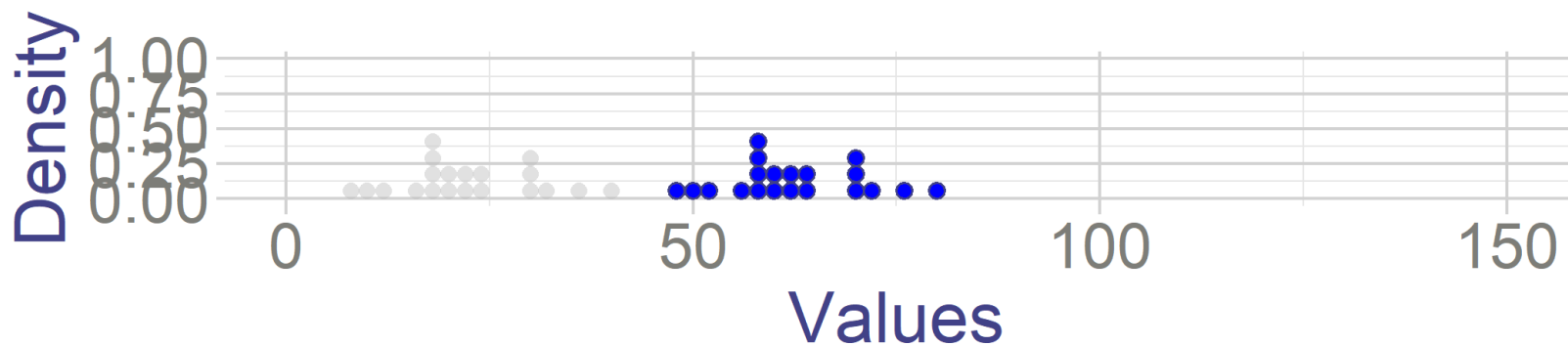
Consider a random variable X with $E(X) = \mu_x$ and standard deviation σ_x .

Ex: X is number of instagram followers distributed like this:



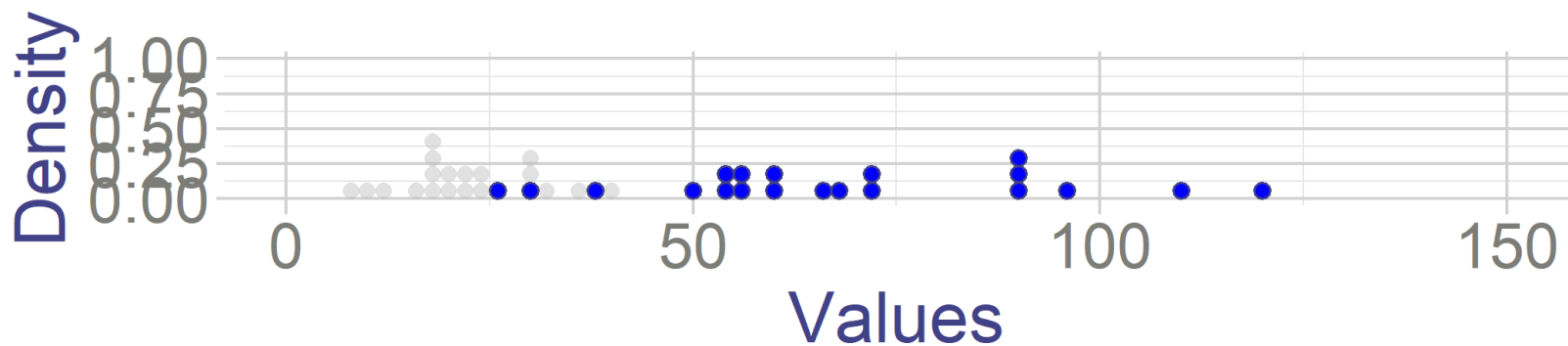
Standard Deviation

- What happens to mean and standard deviation if everyone gets 40 more followers?
- $Y = X + 40$



Standard Deviation

- What happens to mean and standard deviation if everyone followers get multiplied by 3? (without addition)
- $Y = 3 * X$



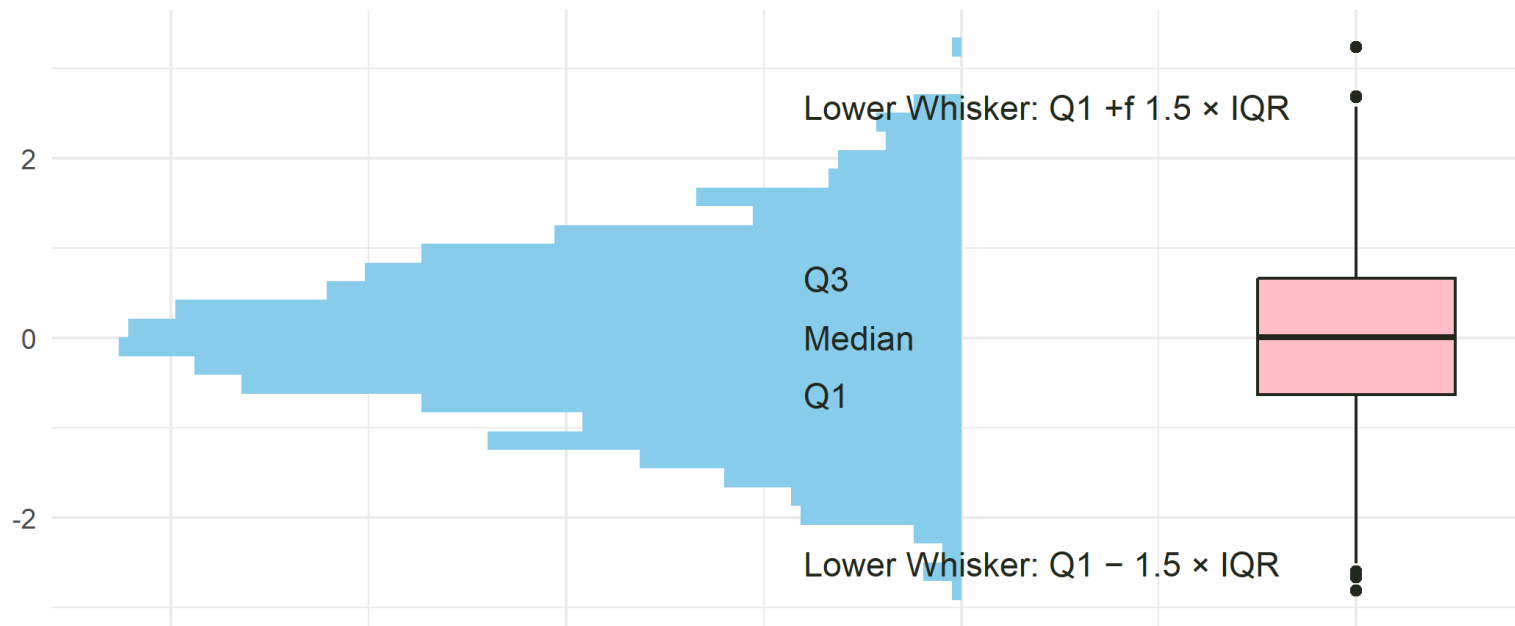
Box and Whiskers plot

- Helps to see the **distribution** of the data and the **outliers**
 - Outliers reveal anomalies and potential data-collection errors
 - A whisker extends at most **$1.5 \times \text{IQR}$** from the box
 - Any point beyond that is flagged as an **outlier**
 - If no point is beyond $1.5 \times \text{IQR}$, the whisker stops at the last datapoint

► **learnr** `W03_learnr.Rmd` → **Exercise 4: Boxplot** — build the five-number summary.

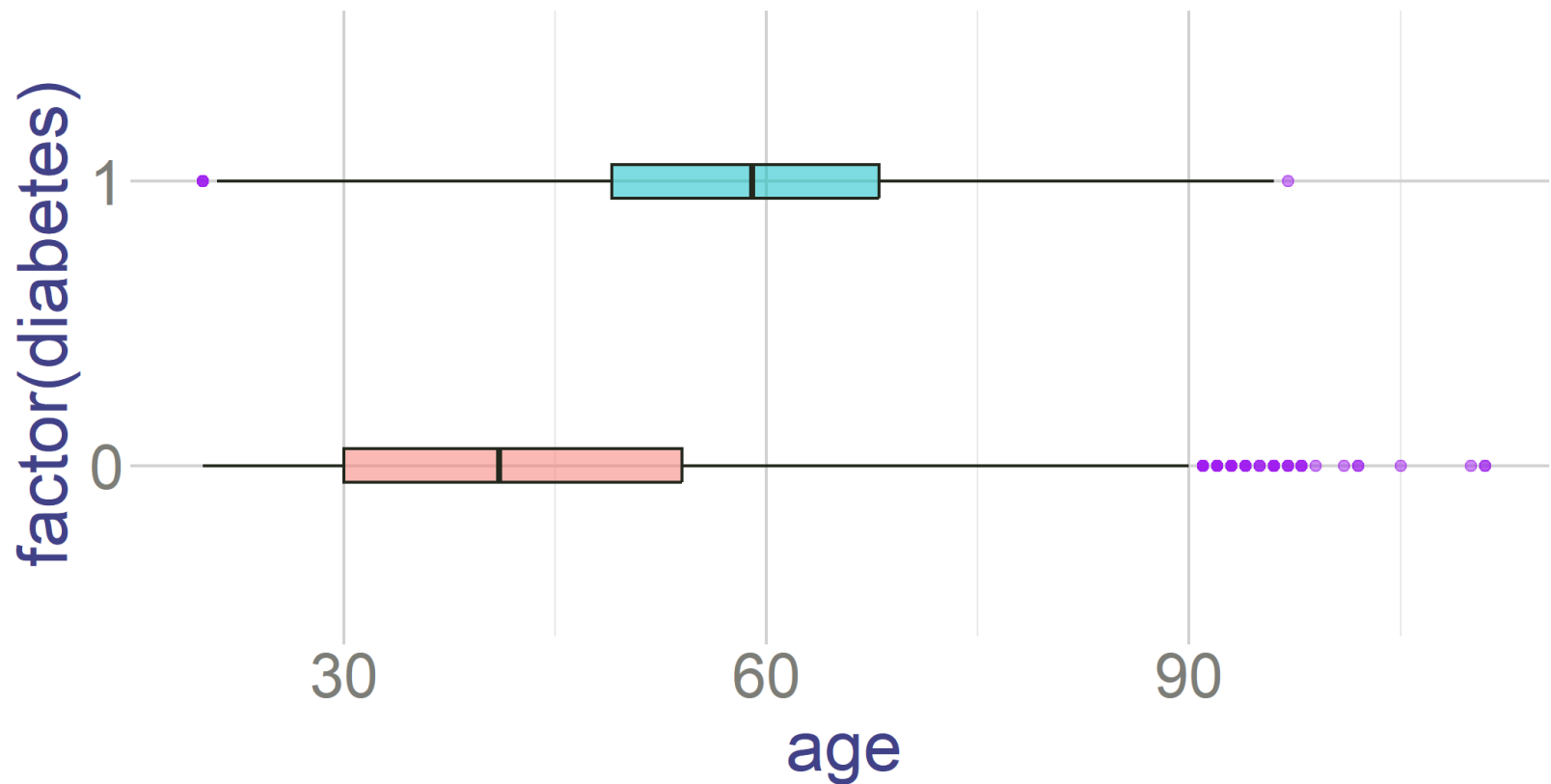
Box and Whiskers plot: anatomy

The box spans Q1–Q3 (the middle 50%); the line inside is the median.



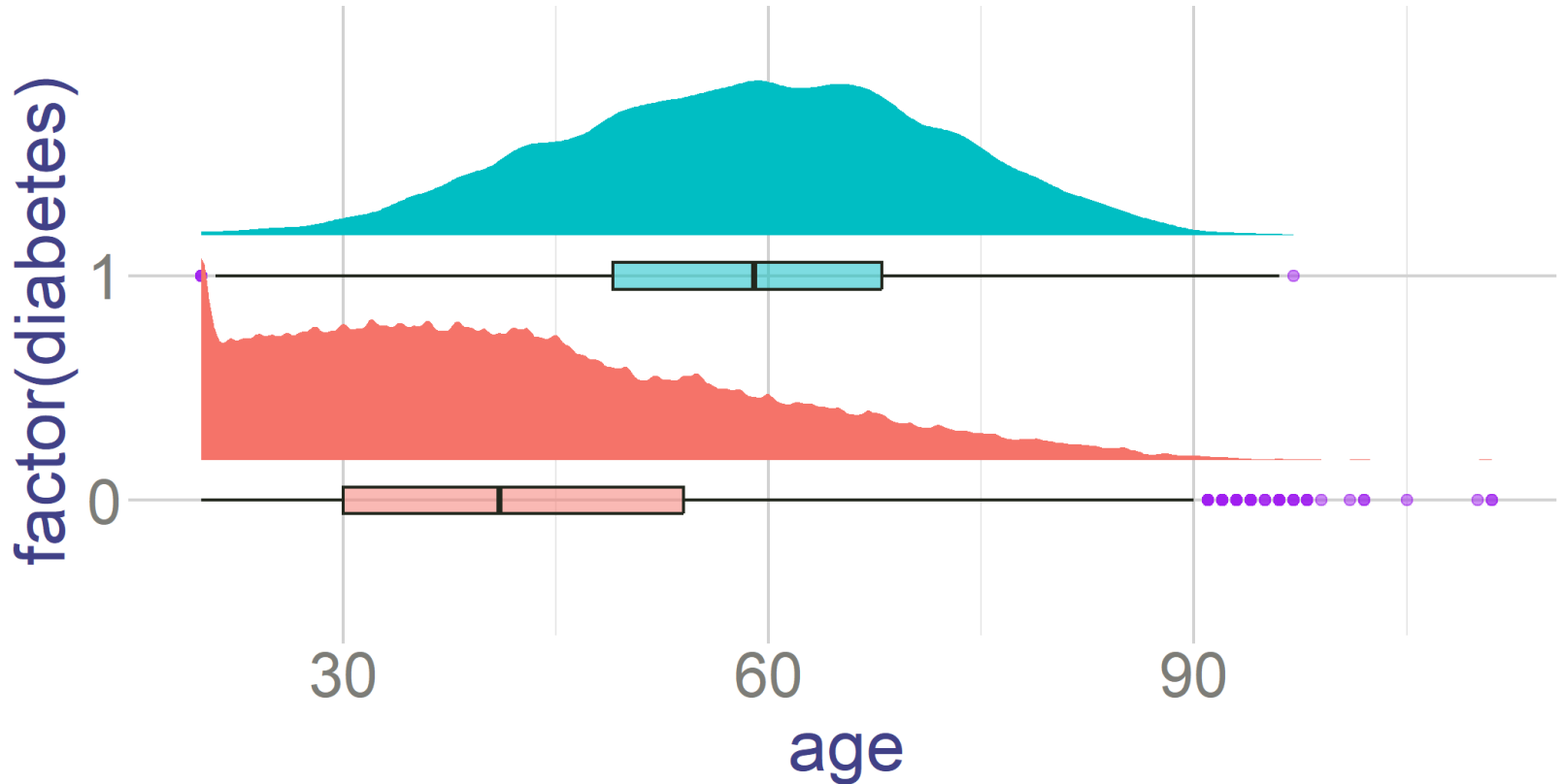
Box and Whiskers plot (cont.)

Dataset comparisons — they summarize data very well (age by diabetes status):



Box and Whiskers plot (cont.)

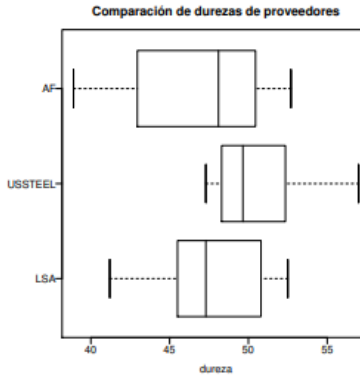
Dataset comparisons — boxplot with the full distribution overlaid (age by diabetes status):



Reading a boxplot (warm-up)

5. [5 puntos] The next plot represents data on the hardness of steel rods for three different suppliers: AF, USSteel, and LSA. Based on this figure, it may be said that:

Hardness of steel rods for three different suppliers



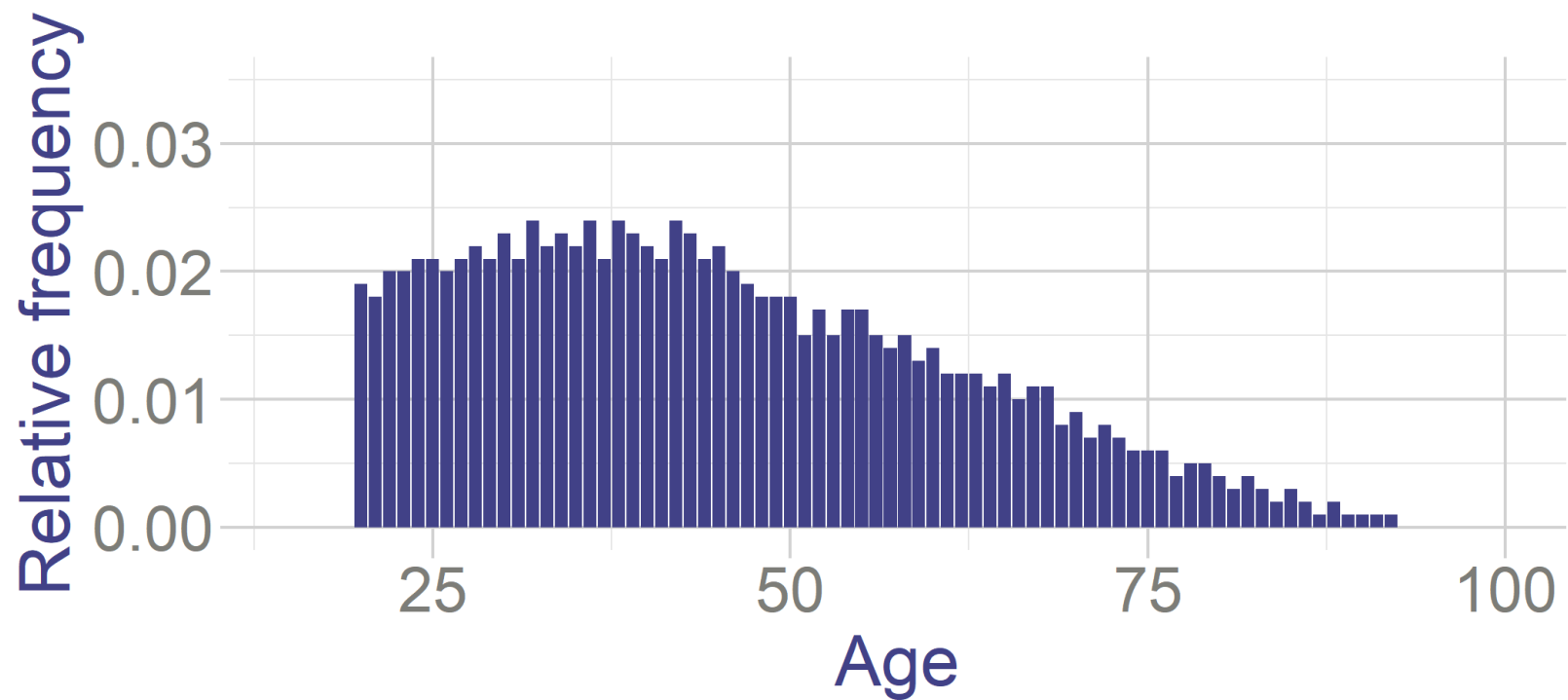
- a) Distributions for all supplier's hardness are skewed to the left.
 - b) AF's hardness seems to have the least dispersion of the three.
 - c) AF's hardness seems to have less dispersion than that of LSA.
 - d) USSTEEL hardness distribution is skewed to the right.
6. [5 puntos] The prior plot representing the hardness of steel rods for three different suppliers, allows us to identify useful:
- a) central tendency and location measures.
 - b) insights about the distribution's dispersion.
 - c) asymmetry among distributions.
 - d) all of the above.

Distributions

How is the data *spread out*? Now picture the *whole* distribution.

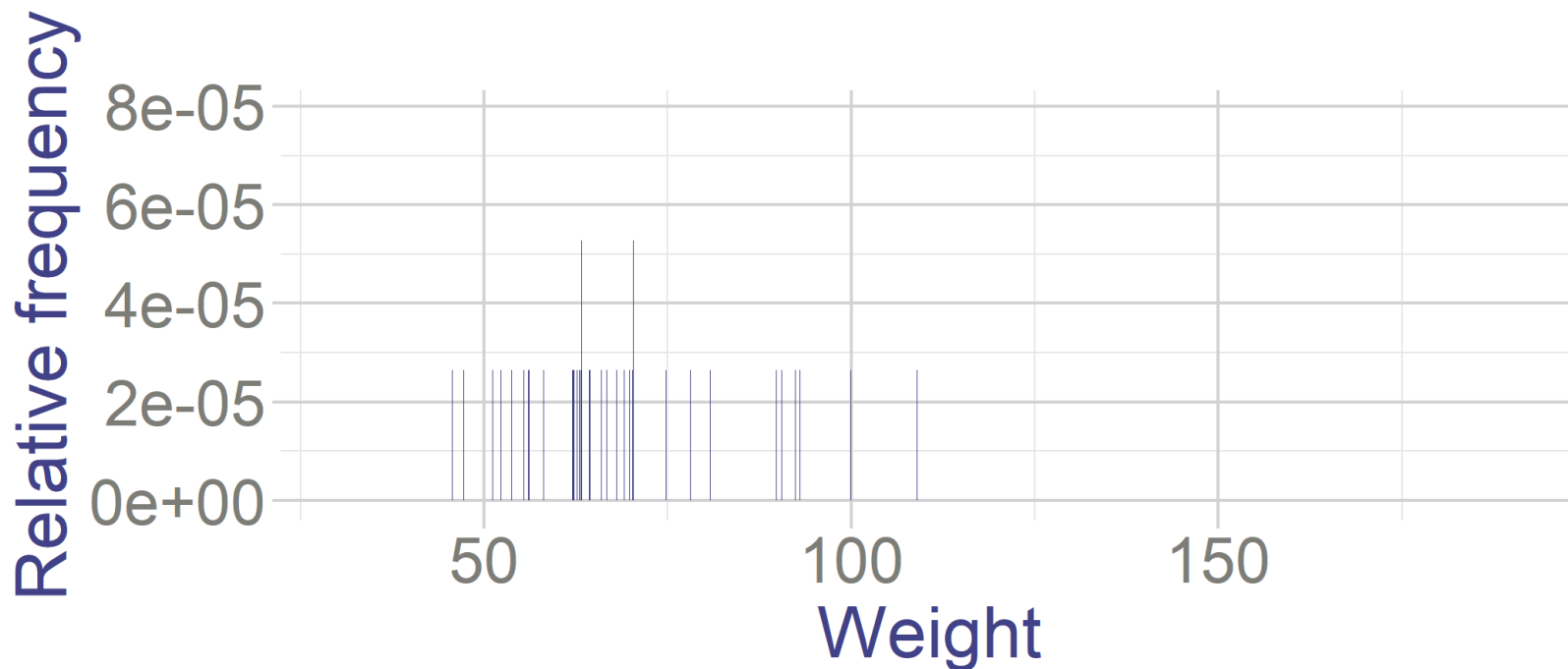
Frequency distribution of a numeric variable

Compare age in the adult population (20+) — how often does each value occur?



Continuous variables: the problem

- What about a **continuous** variable like weight? Why can't we do the same?



- Most values **never repeat**, so each has a tiny relative frequency — the plot is useless.

Histograms

Solution: Group similar values together

- Construct intervals and show how many observations are in a given interval

Process

1. Decide how many intervals
2. And how wide they are
3. Then calculate the absolute and relative frequencies of each interval
4. Plot it with bars

► [learnr W03_learnr.Rmd](#) → **Exercise 6: First histogram** — build it now.

Histograms: choosing bin width

- I want k (example $k=5$) equal intervals
- Divide the **range** of the data (max – min) into k equal intervals

```
max_value <- max(Health_data$weight)
min_value <- min(Health_data$weight)
range <- max_value - min_value
```

```
## [1] "Range= 190.81 - 30.37 = 160.43"
```

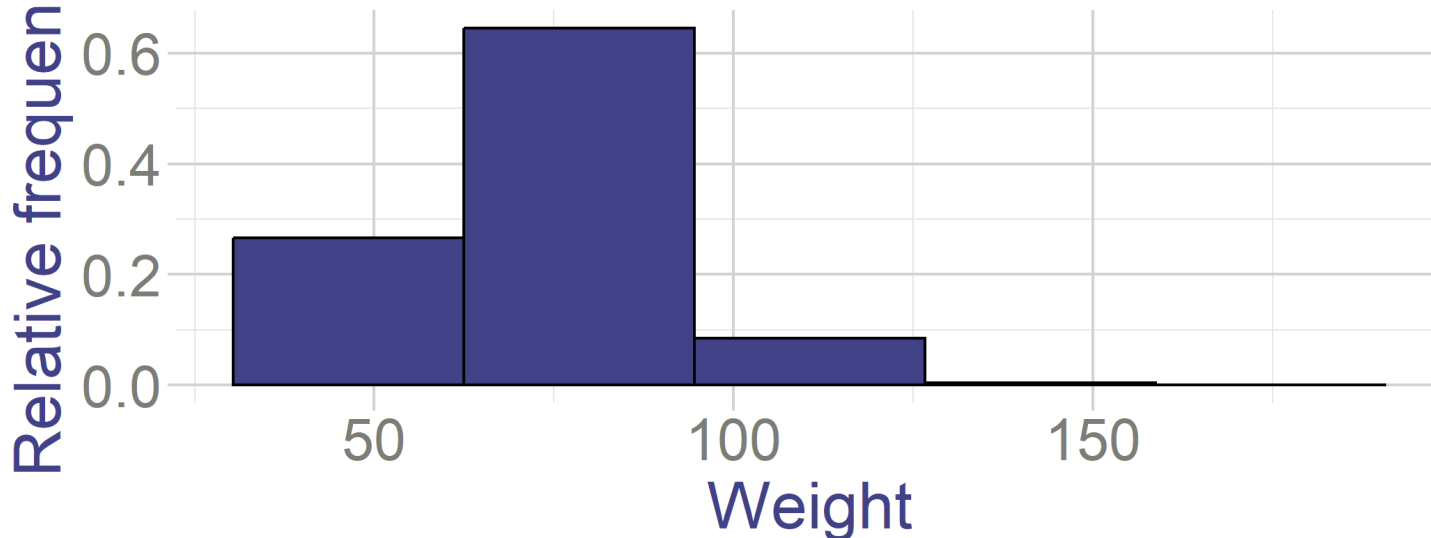
- With 5 intervals, each is ~32 kg wide; the first starts at the min (30.37), the last ends at the max (190.81).

► [learnr W03_learnr.Rmd](#) → **Exercise 7: How many bins?** — bins = 5, 30, 100.

Histogram with 5 classes

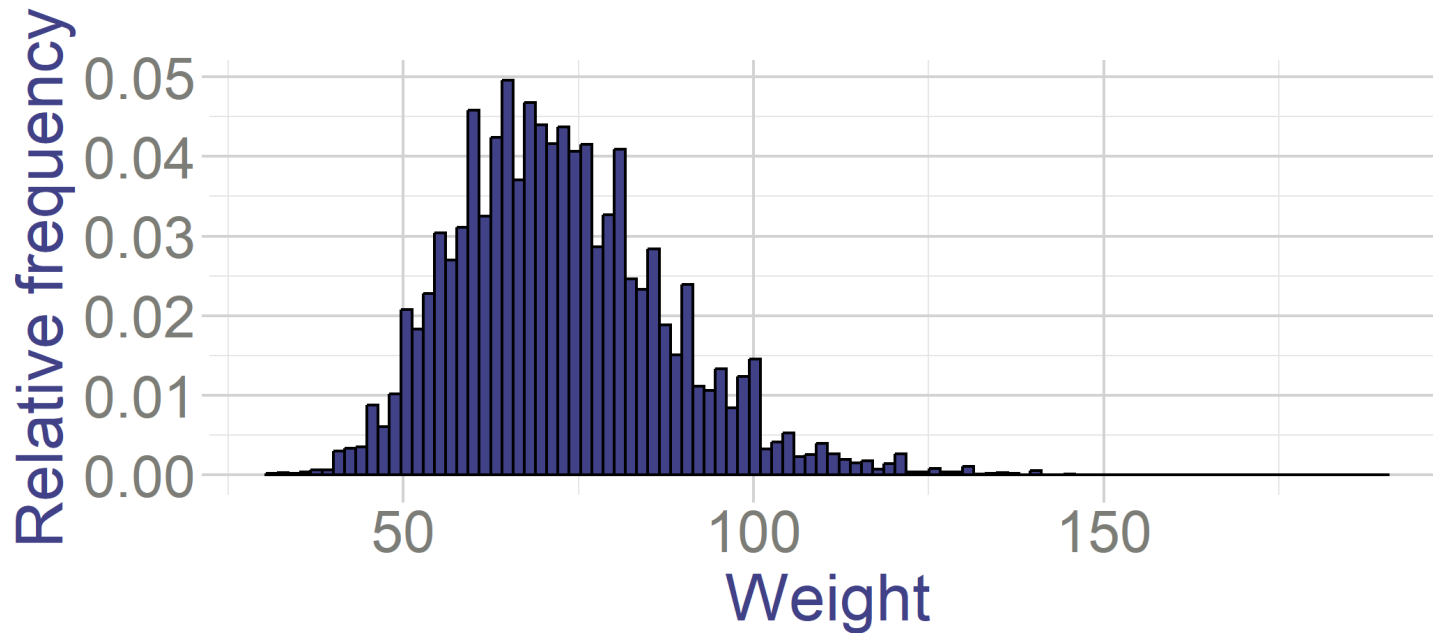
- Midpoint = center of the interval (center of the bar)
- P_i = cumulative frequency: share of observations in this or a smaller interval
 - Example: $P_{(62.46-94.55)} = 0.911 \rightarrow 91.1\%$ of people weigh less than 94.55 kg

Histogram with 5 Classes



► `learnr W03_learnr.Rmd` → Exercise 8: Add a density curve.

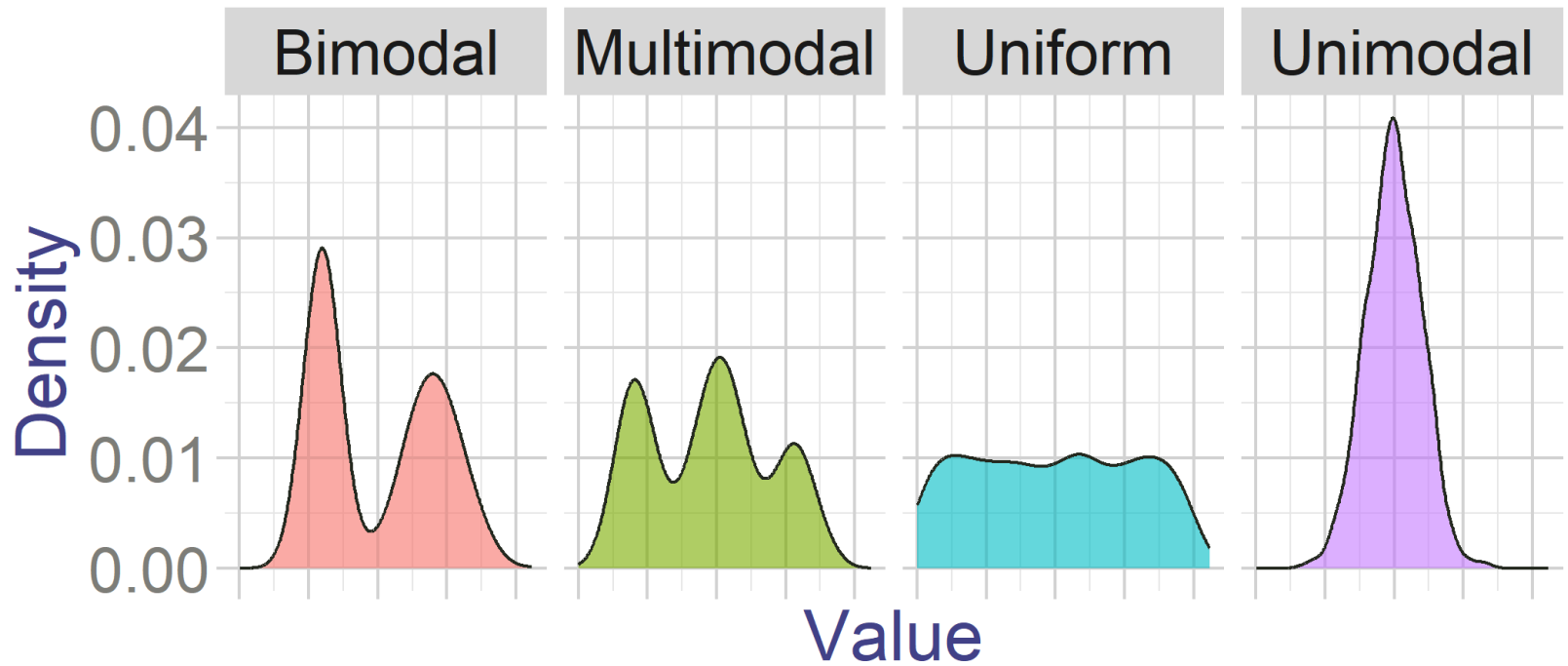
Histogram with 100 classes



- Helps to see the distribution and outliers — but is more always better?
- With smaller and smaller intervals, the histogram tends to the **probability density function**.

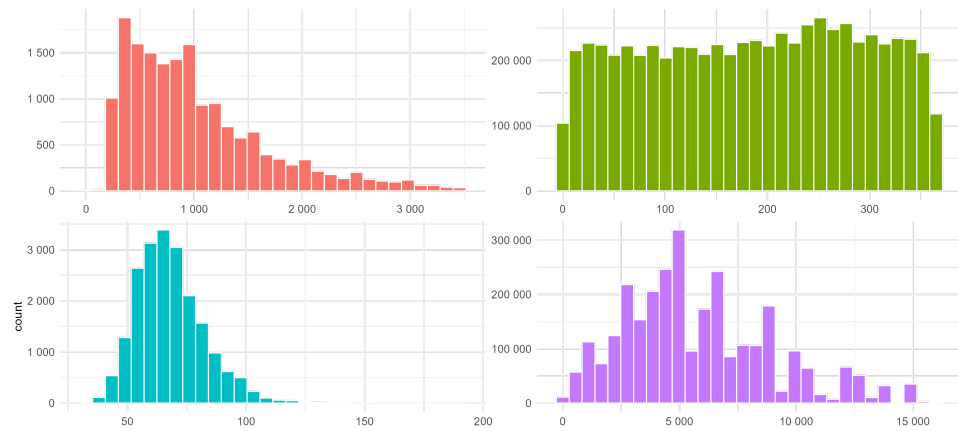
Distribution Shapes: Modality

How many "peaks" does the distribution have?



Which is uniformly distributed?

1. Weights of Adult Females
2. Salaries in Mexico
3. Airbnb prices in CDMX
4. Birthdays of Classmates (day of the month)



Value

Distribution Shapes: Skewness

Is the tail longer on the left or the right?



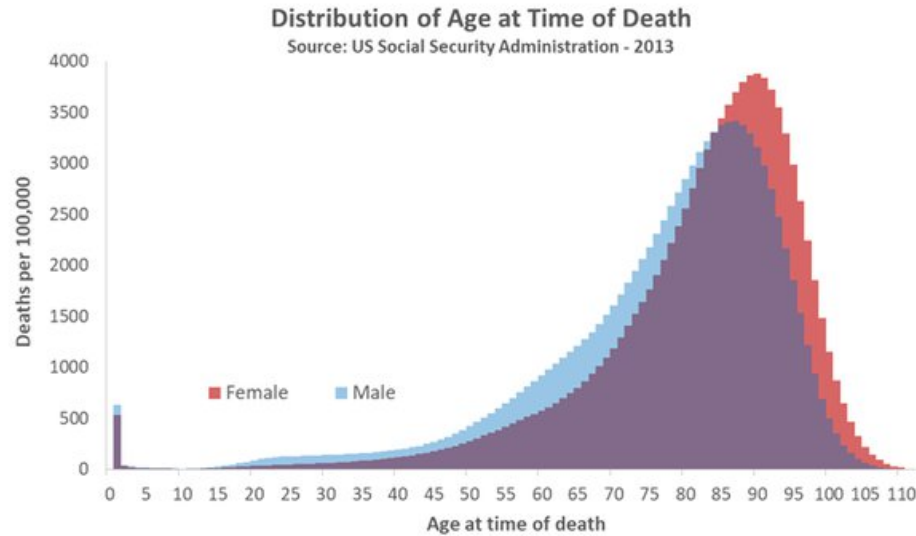
► `learnr W03_learnr.Rmd` → Exercise 9: Shape — mean vs median.

Shape in R: the analyst's move

- The whole job of reading a distribution: **plot it, then name its shape.**
- **Modality** — count the peaks: one (unimodal), two (bimodal), many.
- **Skewness** — which tail is longer: left, right, or symmetric.
- In R: `ggplot(df, aes(x = var)) + geom_histogram()`, then eyeball it.
- That's enough to answer most business questions — do it, don't derive it.

► [learnr W03_learnr.Rmd](#) → **Exercise 5: Compare groups with boxplots.**

Age at death (a left-skewed variable)



Exam Spotlight

The real questions — and how we solve them.

EXAM QUESTION — Fall 2025 Midterm 1

Real question from last year's Midterm 1 — **Households & Environment Survey** (each household surveyed once, 2017). **We solve this together in class.**

Q3 (Distributions). *An energy company wants to understand the distribution of weekly household gasoline spending to design a fuel discount program.*

- a) **Visualize** the distribution of `gasoline_spending_weekly`.
- b) **Describe its shape** — modality and skewness.

Pick the right plot (a **histogram**), then read the **shape** straight off it: one peak or several, and which tail is longer.

Practice now (this week's pack): the histogram + shape-description problems in the practice set.

Q5 (Boxplot interpretation). The boxplot compares **monthly electricity bills** for households that use electricity for cooling (AC/fans) vs those that do not.

- a) Which group has the **higher median** electricity bill?
- b) Which group shows **greater variability (spread)**?

Read it straight off the five-number summary: **median line** for (a), **box width / whisker length** for (b).

Practice now (this week's pack): the boxplot problems in the practice set.

What you must be able to do this week

- **Read a histogram** and describe the distribution's **shape** — modality (uni-/bi-/multimodal) and skewness (which tail is longer).
- Compute and interpret **range, IQR, variance, and standard deviation**, and know how each reacts to a linear change $y = bx + a$ and which one an **outlier** moves.
- Choose the right **spread measure** for the question — robust (IQR) vs sensitive to outliers (range, variance).
- Read a **boxplot's five-number summary** — median, quartiles, whiskers, and flagged outliers.

Next week (C3): from describing data to reasoning about **uncertainty** — **estimators and the Central Limit Theorem (CLT)**.

Appendix — optional (extra & theory)

Variance & Standard Deviation — why square then root?

- Why do we first take squares and then take square root?
- Can't we just do $\frac{1}{N} \sum_{i=1}^N (x_i - \mu)$?
- NO! Because $\sum_{i=1}^N (x_i - \mu) = 0$

Variance & Standard Deviation (cont.) — why not MAD?

- Why don't we just do Mean Absolute Deviation?

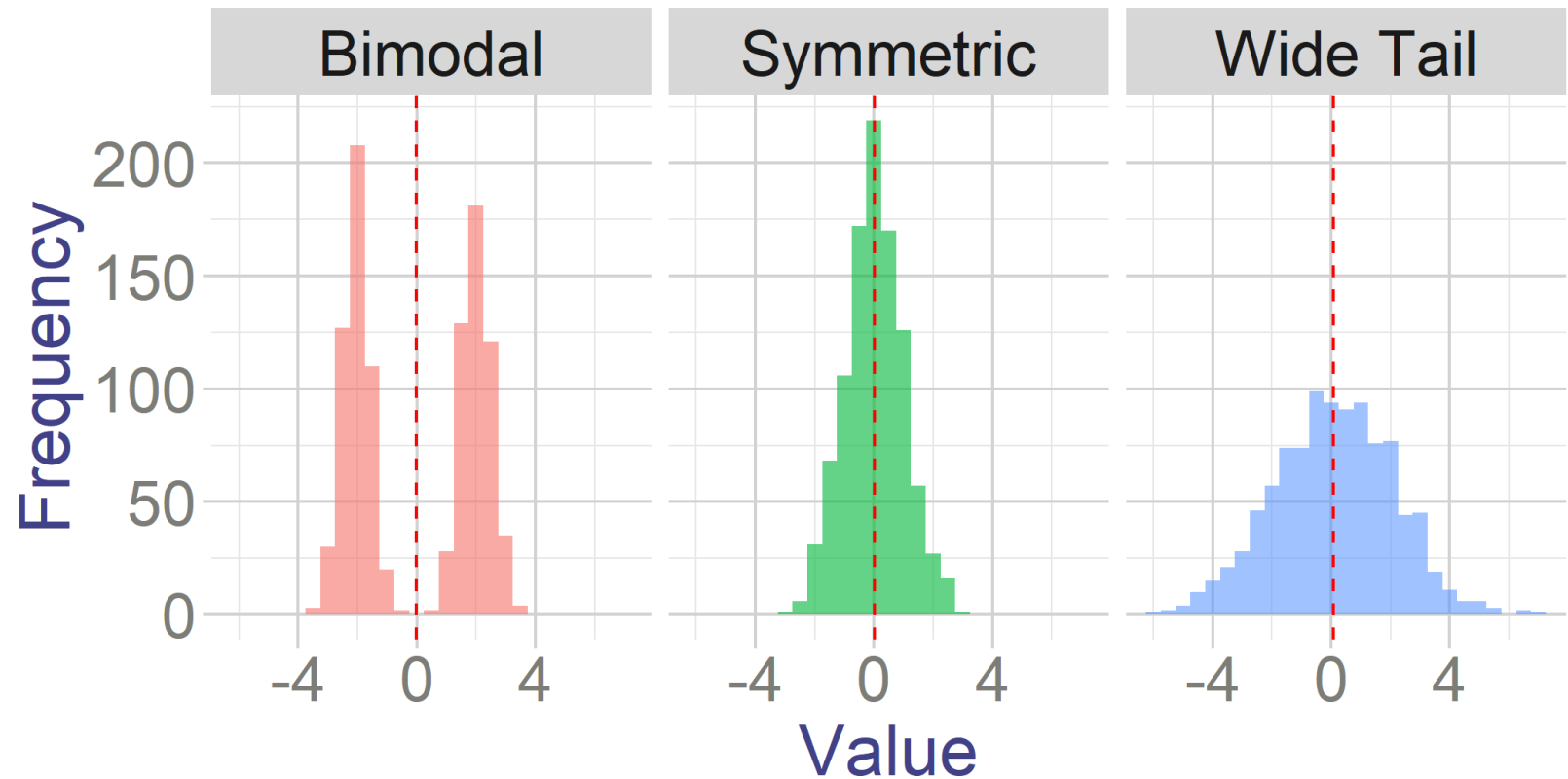
- $MAD = \frac{1}{N} \sum_{i=1}^N |(x_i - \mu)|$

- MAD is not differentiable at 0 :(
- Variance puts more weight on far away observations.
- It's a weighted average distance, where weights are distances themselves.

$$\sigma^2 = \sum_{i=1}^N (x_i - \mu)^2 = \frac{1}{N} \sum_{i=1}^N \underbrace{|(x_i - \mu)|}_{\text{Weight}} * \underbrace{|(x_i - \mu)|}_{\text{Distance}}$$

- It makes it very sensitive to outliers, which get a lot of weight!
- Standard deviation retains this property

Variance: same center, different spread (no annotation)



Coefficient of Variation

Coefficient of Variation divides the standard deviation by the mean.

$$C.V. = \frac{\sigma}{|\mu|}$$

And sample equivalent

$$c.v. = \frac{s}{|\bar{x}|}$$

- Why?
 - It expresses standard deviation as proportion of the mean
 - Small value means variation is low compared to the mean
 - It is unit free
 - You can compare it across variables with different units/magnitudes

Coefficient of Variation

Example - variation of stocks in different currencies

Show

6

 entries

Date	MXN_Stock	USD_Stock
2023-07-01	91.59	1.01
2023-07-02	96.55	1.16
2023-07-03	123.38	1.02
2023-07-04	101.06	1.07
2023-07-05	101.94	1.09
2023-07-06	125.73	0.9

Showing 1 to 6 of 20 entries

Previous

1

2

3

4

Next

- **Standard deviation:**
 - USD: 0.149
 - MXN: 14.59
- **Coefficient of variation:**
 - USD: 0.12
 - MXN: 0.14

Coefficient of Variation

So more generally, if $y_i = bx_i$, then

$$C.V._y = \frac{\sigma_y}{|\mu_y|} = \frac{|b|\sigma_x}{|b\mu_x|} = C.V._x$$

What if $y_i = bx_i + a$?

Then

$$C.V._y = \frac{\sigma_y}{|\mu_y|} = \frac{|b|\sigma_x}{|b\mu_x + a|} \neq C.V._x$$

Standard Deviation (cont.) – $Y = X + 40$

$$\mu_Y = E(Y) = \frac{\sum_{i=1}^N (y_i + 40)}{N} = \frac{\sum_{i=1}^N (x_i + 40)}{N} = \frac{\sum_{i=1}^N x_i + N * 40}{N} = E(X) + 40 = \mu_X + 40$$

$$\sigma_Y = \sqrt{Var(Y)} = \sqrt{\frac{\sum_i (y_i - \mu_y)^2}{N}} = \sqrt{\frac{\sum_i ((x_i + 40) - (\mu_x + 40))^2}{N}} = \sqrt{\frac{\sum_i (x_i - \mu_x)^2}{N}} = \sqrt{Var(X)} = \sigma_X$$

$$E(X + c) = E(X) + c \quad \text{and} \quad Var(X + c) = Var(X)$$

Standard Deviation (cont.) – $Y = 3X$

$$\mu_Y = E(Y) = \frac{\sum_i y_i}{N} = \frac{\sum_i 3x_i}{N} = 3 \frac{\sum_i x_i}{N} = 3E(X) = 3\mu_X$$

$$\sigma_Y = \sqrt{Var(Y)} = \sqrt{\frac{\sum_i (y_i - \mu_y)^2}{N}} = \sqrt{\frac{\sum_i (3x_i - 3\mu_x)^2}{N}} = \sqrt{3^2 \frac{\sum_i (x_i - \mu_x)^2}{N}} = \sqrt{3^2 Var(X)} = 3\sigma_X$$

$$E(cX) = cE(X) \quad \text{and} \quad Var(cX) = c^2 Var(X)$$

Q4 (Associations). *A transport consultant needs to know what drives gasoline spending.* Which is more strongly associated with **weekly gasoline spending** — the **number of vehicles** or the **household size**? Choose an appropriate metric, compute it, and interpret.

Would the conclusion change if spending were in USD instead of MXN (1 USD = 17 MXN)?

This is a **correlation vs covariance** question: pick the metric, compute both associations, compare.

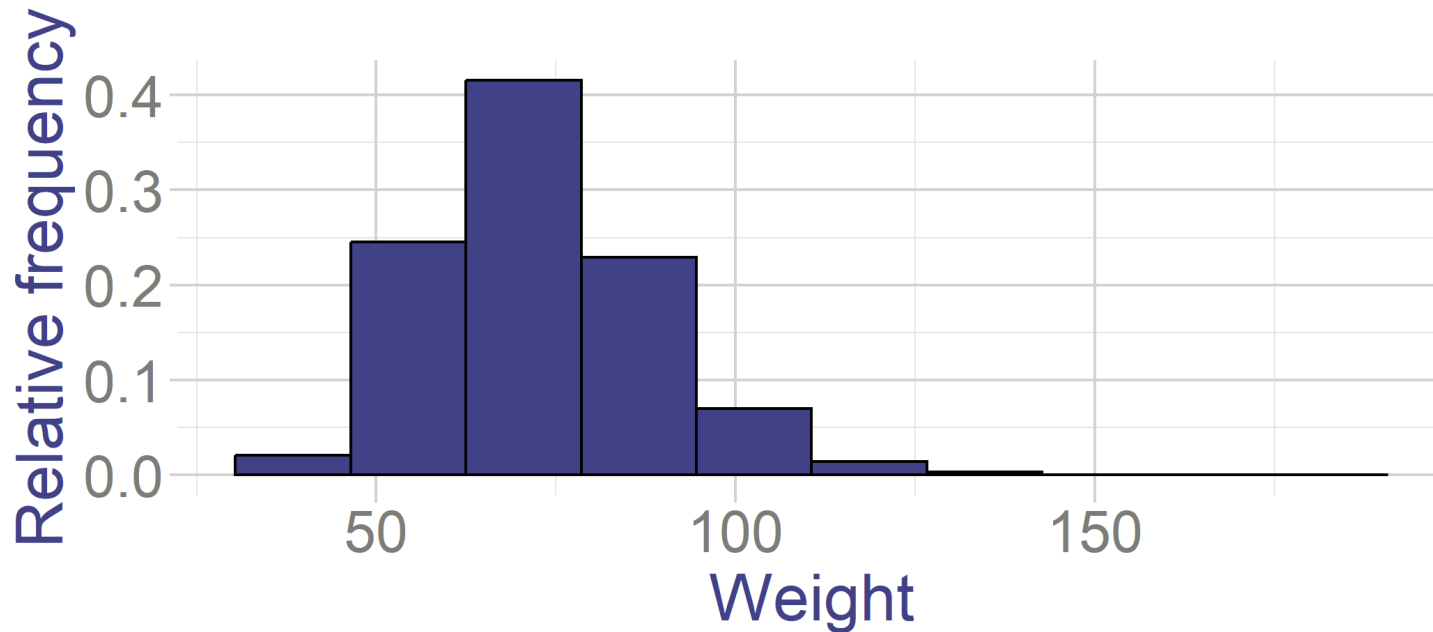
► **learnr** `W03_learnr.Rmd` → **Exercises 1–3: Range, IQR, variance & SD.**

Histogram with 5 classes: the table

Interval	Midpoint	n _i	p _i	P _i
30.37 - 62.46	46.42	10068	0.266	0.266
62.46 - 94.55	78.50	24430	0.645	0.911
94.55 - 126.63	110.59	3206	0.085	0.996
126.63 - 158.72	142.68	143	0.004	1.000
158.72 - 190.81	174.76	11	0.000	1.000

Histogram with 10 classes

More, narrower bins reveal more detail.



Histogram with 10 classes: the table

Show

6

 entries

Interval	Midpoint	n_i	p_i	P_i
30.37 - 46.42	38.4	796	0.021	0.021
46.42 - 62.46	54.44	9272	0.245	0.266
62.46 - 78.5	70.48	15742	0.416	0.682
78.5 - 94.55	86.53	8688	0.229	0.911
94.55 - 110.59	102.57	2661	0.07	0.982
110.59 - 126.63	118.61	545	0.014	0.996

Showing 1 to 6 of 10 entries

Histogram with 100 classes: the table

Show 6 entries

Interval	Midpoint	n_i	p_i	P_i
30.37 - 31.98	31.18	8	0	0
31.98 - 33.58	32.78	11	0	0.001
33.58 - 35.19	34.39	7	0	0.001
35.19 - 36.79	35.99	16	0	0.001
36.79 - 38.4	37.59	24	0.001	0.002
38.4 - 40	39.2	24	0.001	0.002

Showing 1 to 6 of 100 entries

Probability Density Function (PDF)

- The **pdf** describes the probability distribution of a **continuous** random variable.
- It does **not** give the probability at a single value (always 0 for a continuous variable)
- It shows in which intervals the variable most often appears
- Used to compute the probability of the variable falling in an interval
- **Area under it always adds up to 1**

Example: X = weight of adults in Mexico. The PDF describes how likely a weight is to fall in a range (e.g. between 58 and 60 kg).

PDF: computing a probability over a range

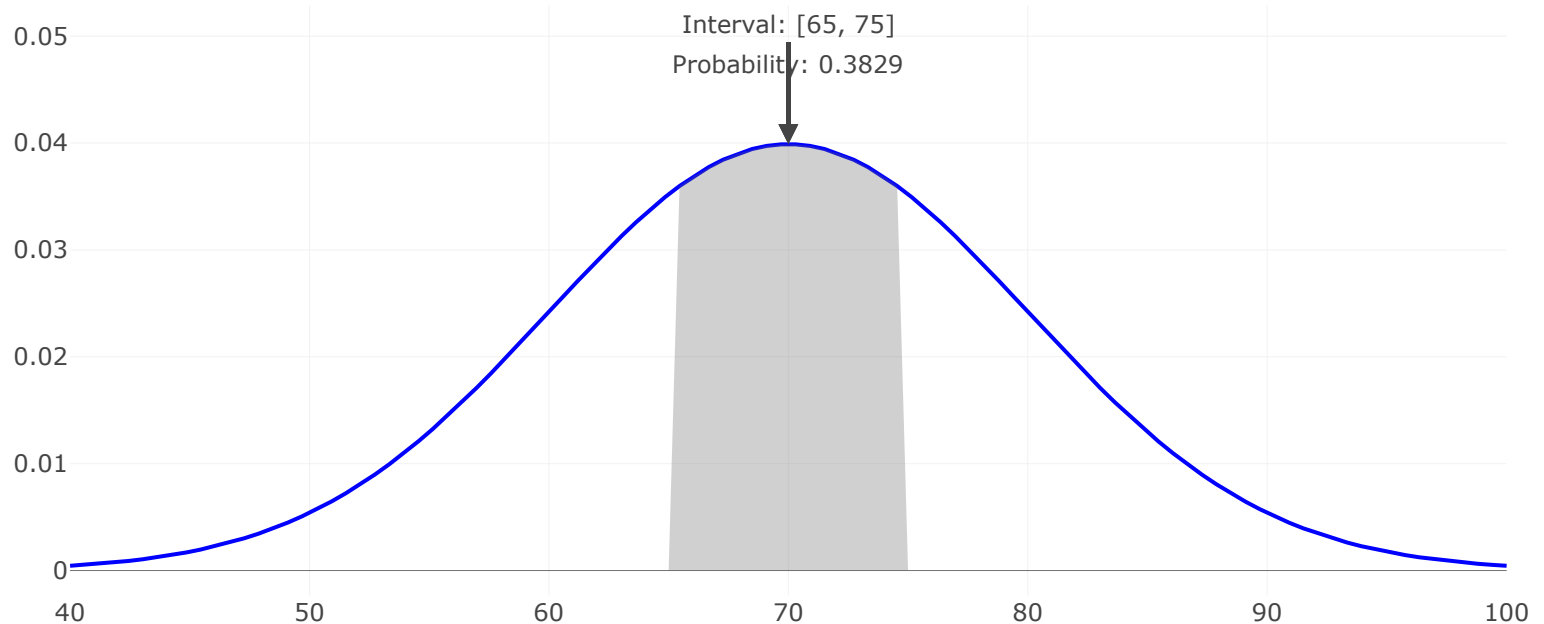
To get the probability that X falls in $[a, b]$, integrate the PDF from a to b :

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

What share of the population weighs between **65 and 75 kg**?

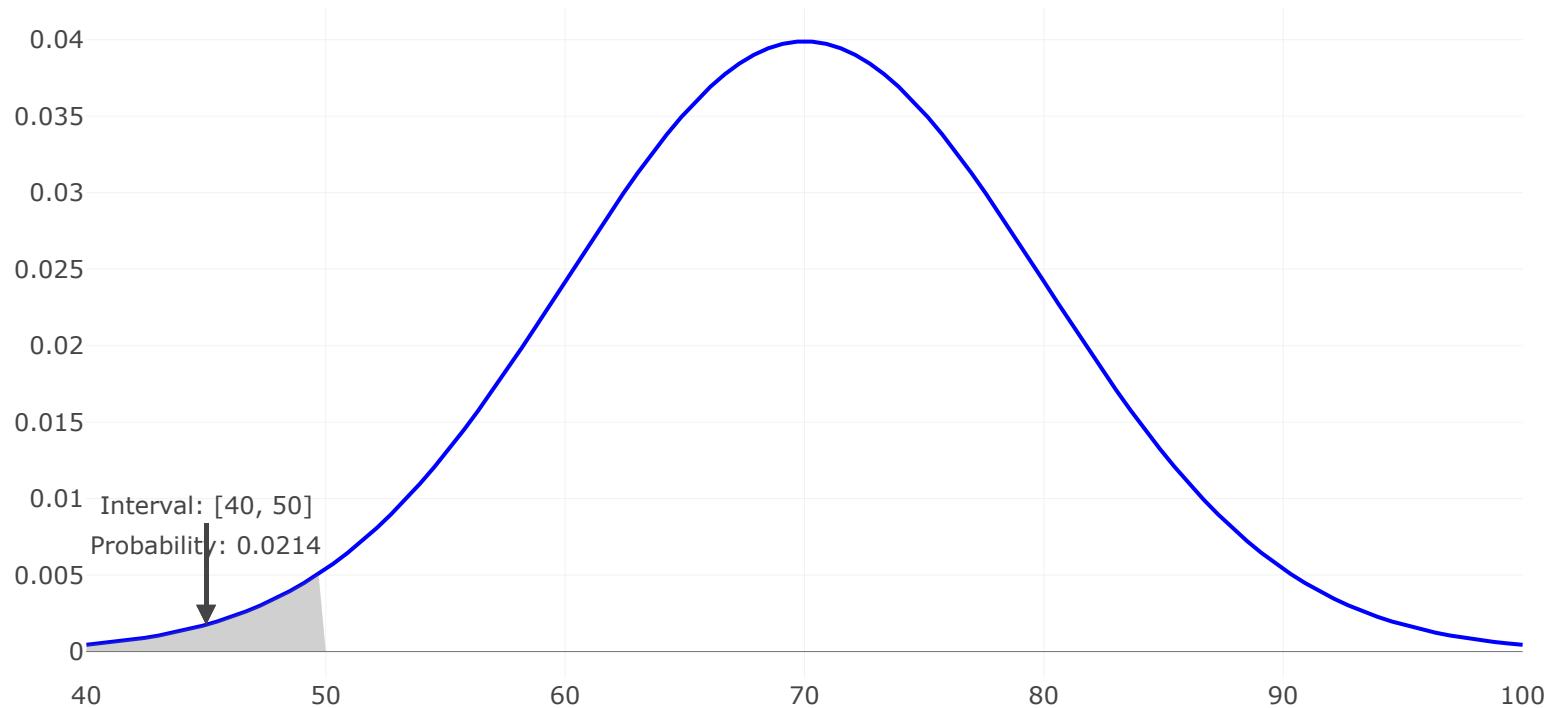
PDF: computing a probability over a range (cont.)

What share of the population weighs between **65 and 75 kg**? The shaded area under the PDF (interactive).



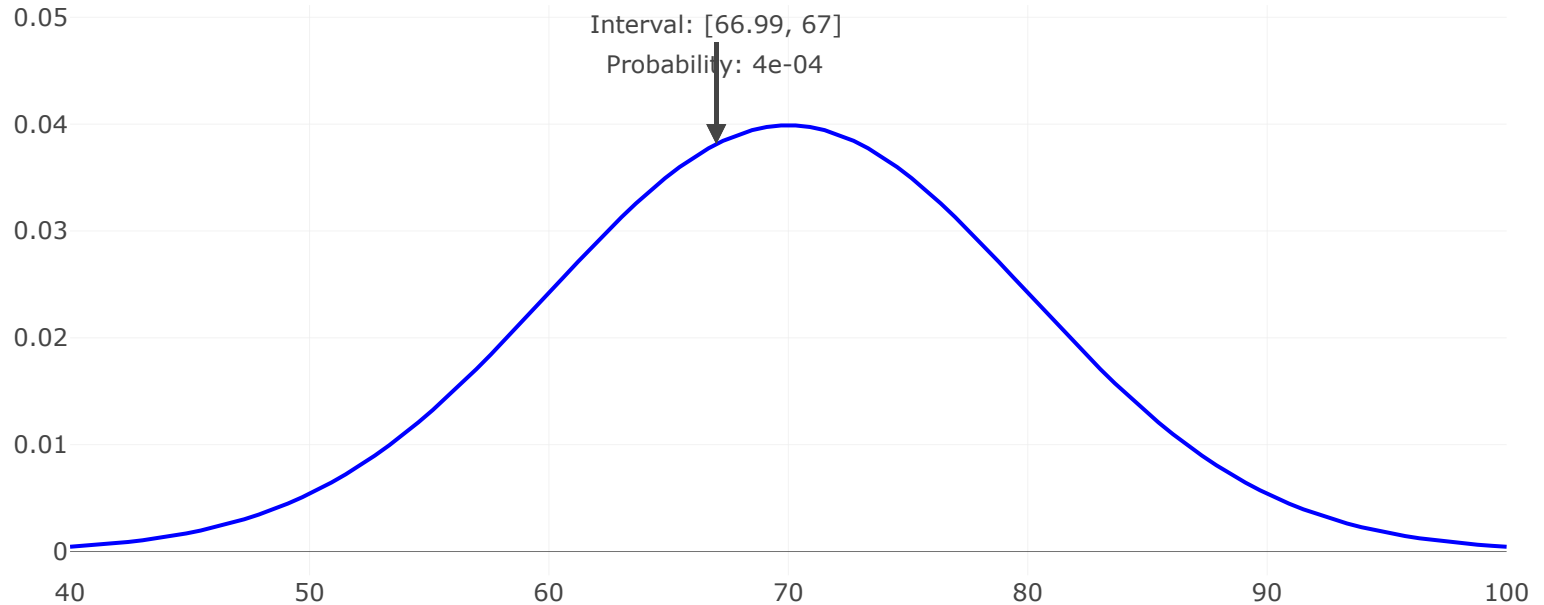
PDF: a wider range

What share of the population weighs between **40 and 50 kg**?



PDF: a razor-thin range

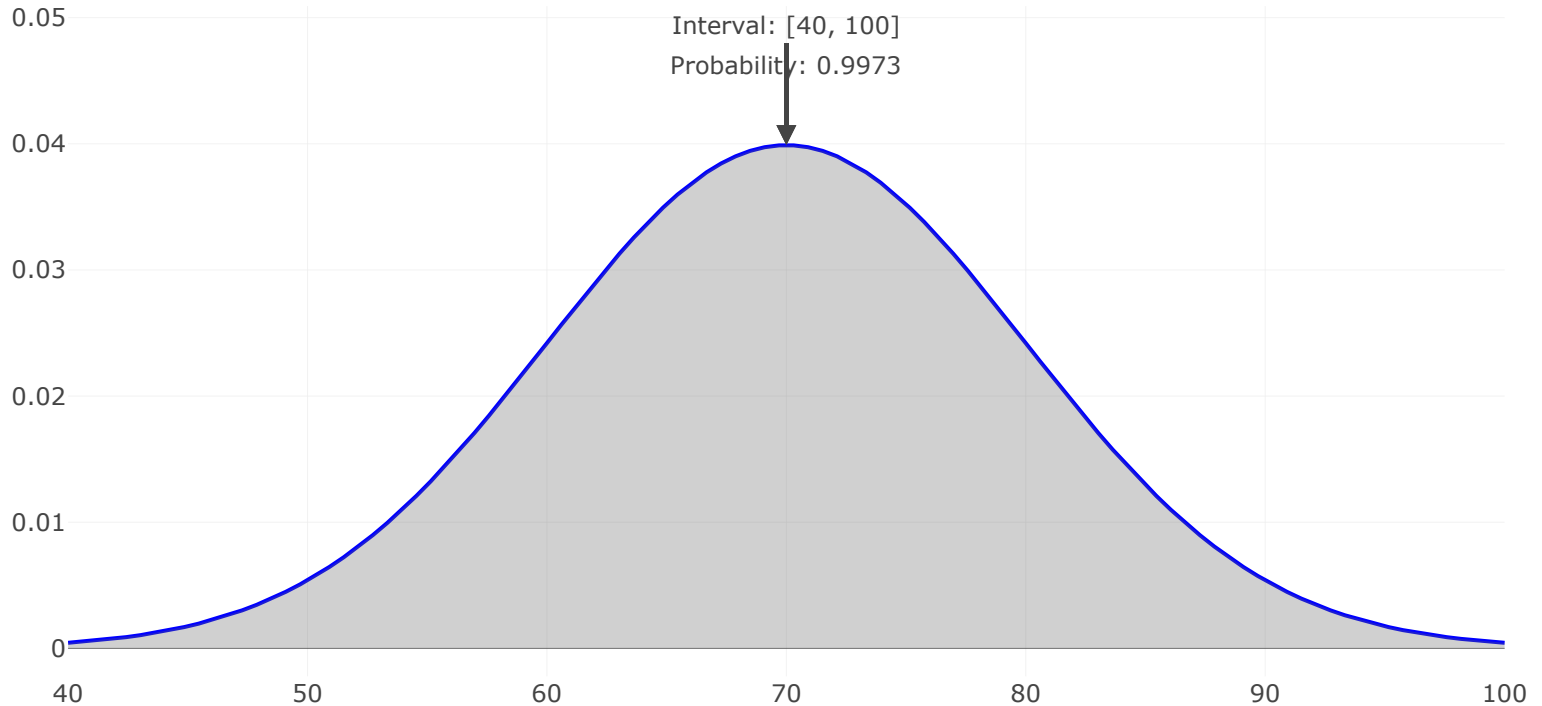
What share of the population weighs between **66.99** and **67 kg**?



A single point has ~ 0 probability — that's why the pdf gives **densities**, not probabilities.

PDF: the whole range integrates to 1

What share of the population weighs between **40 and 100 kg**? (almost everyone)



- The area under the **entire** curve is 1.

What if we want the proportion of people who weigh 50 kg or less?

We need the *cumulative* picture — enter the CDF.

Cumulative Distribution Function (CDF)

The **Cumulative Distribution Function** (CDF) gives the probability that X takes a value **less than or equal to** a specific value.

For a continuous X with PDF $f(x)$, the CDF is:

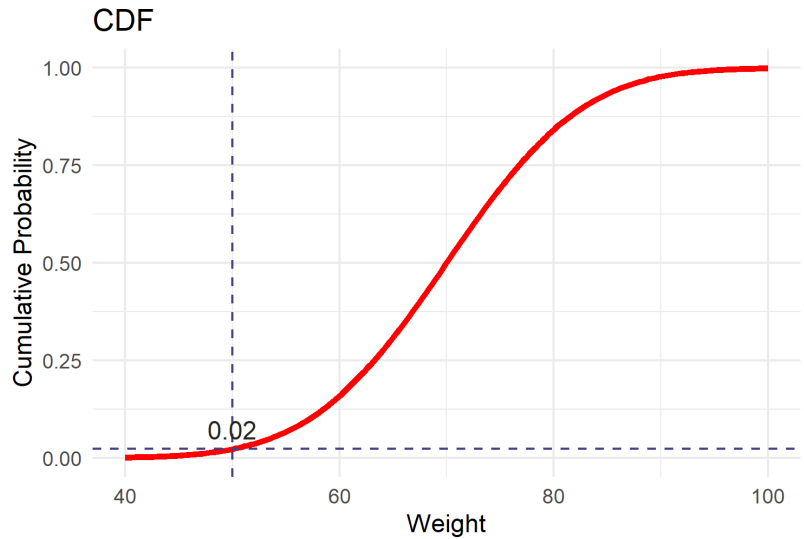
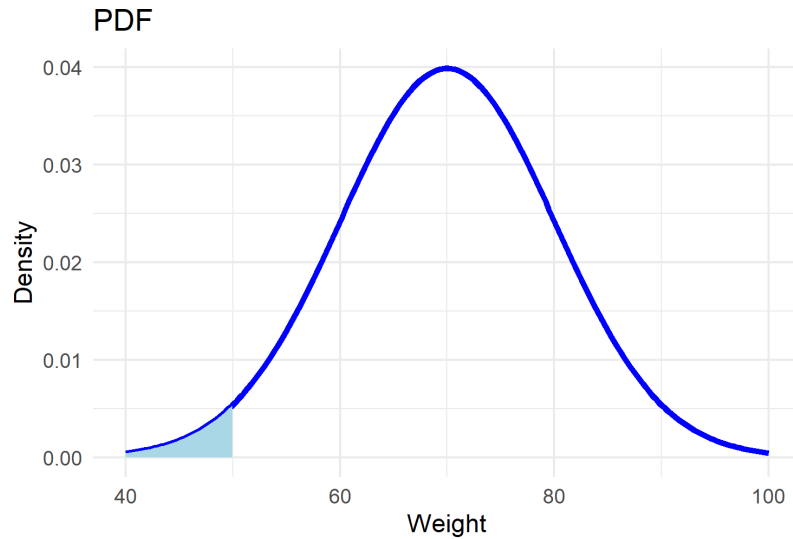
$$F(x) = \int_{-\infty}^x f(t) dt = P(X \leq x)$$

Characteristics:

- Starts at 0 (as $x \rightarrow -\infty$) and approaches 1 (as $x \rightarrow +\infty$)
- Non-decreasing and right-continuous

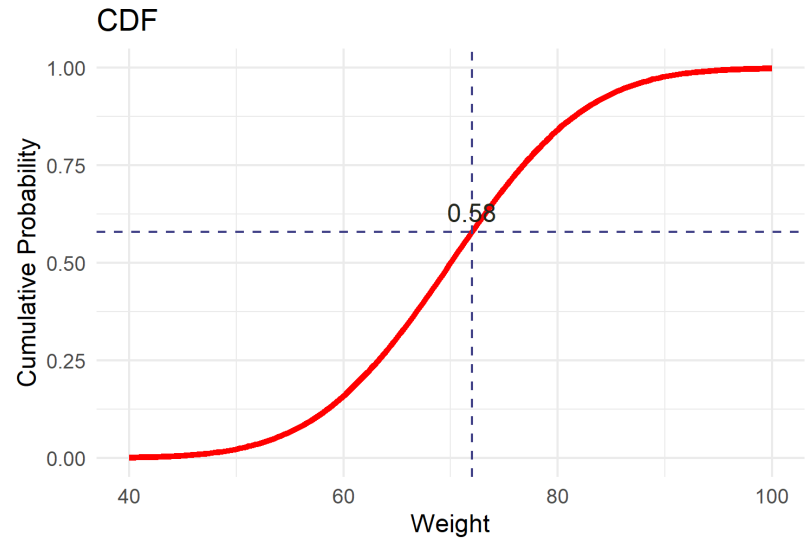
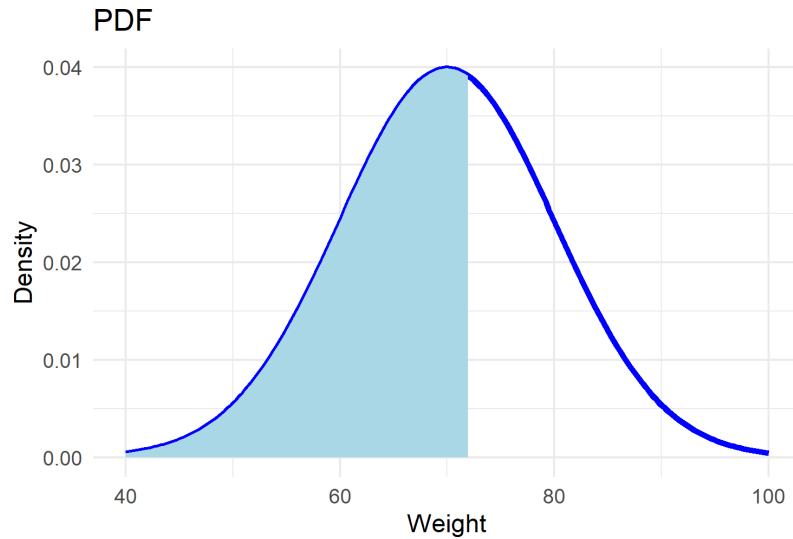
CDF Example 1: weight in the population

$$F(50) = P(X \leq 50) = 0.02$$



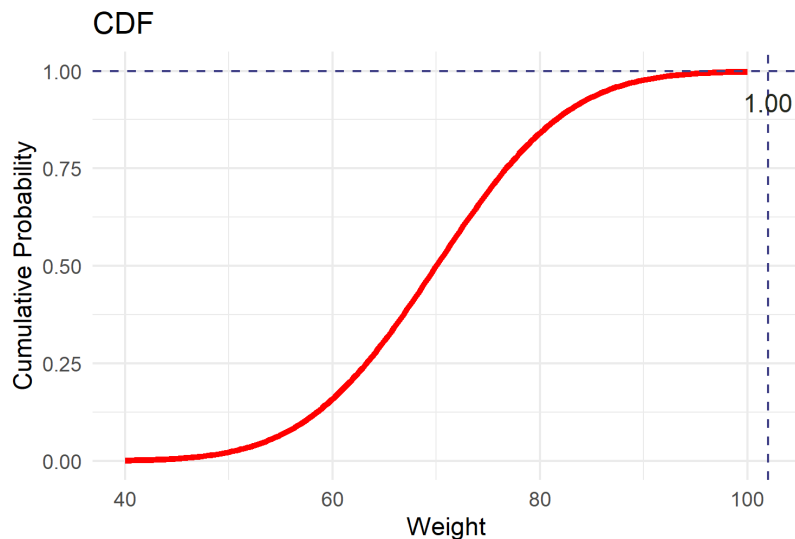
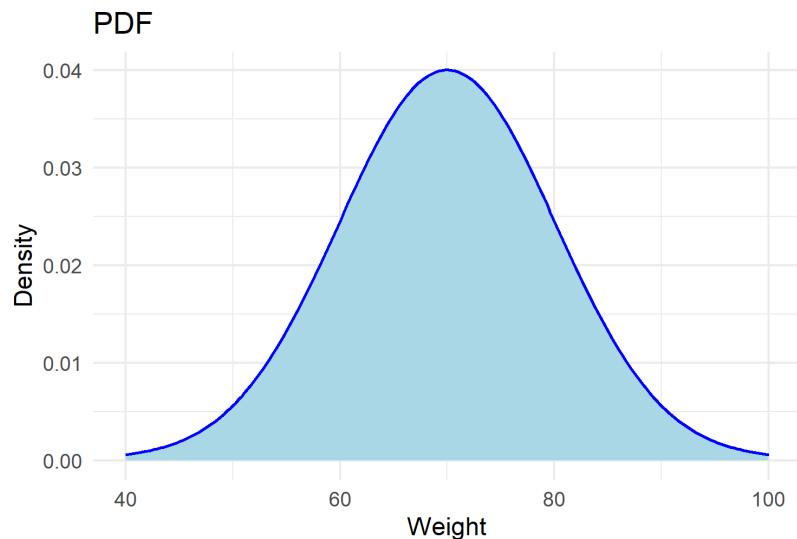
CDF Example 2: weight in the population

$$F(72) = P(X \leq 72) = 0.58$$



CDF Example 3: weight in the population

$$F(102) = P(X \leq 102) = 0.99$$

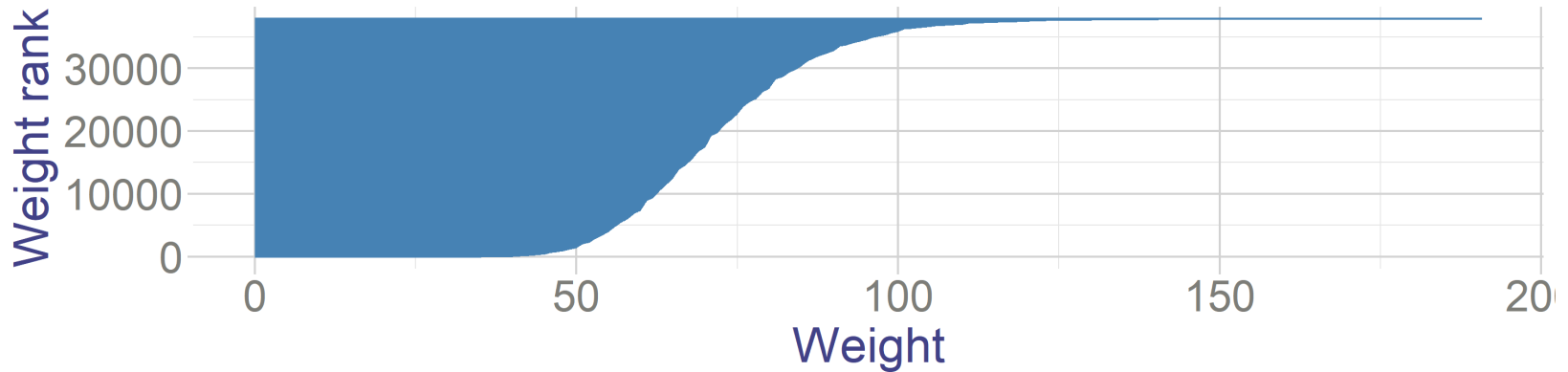


Never integrate a CDF! (it is already the integral of the pdf)

Empirical CDF

What if we only have a **sample** and don't know the true pdf? Line up everyone's weight and sort it.

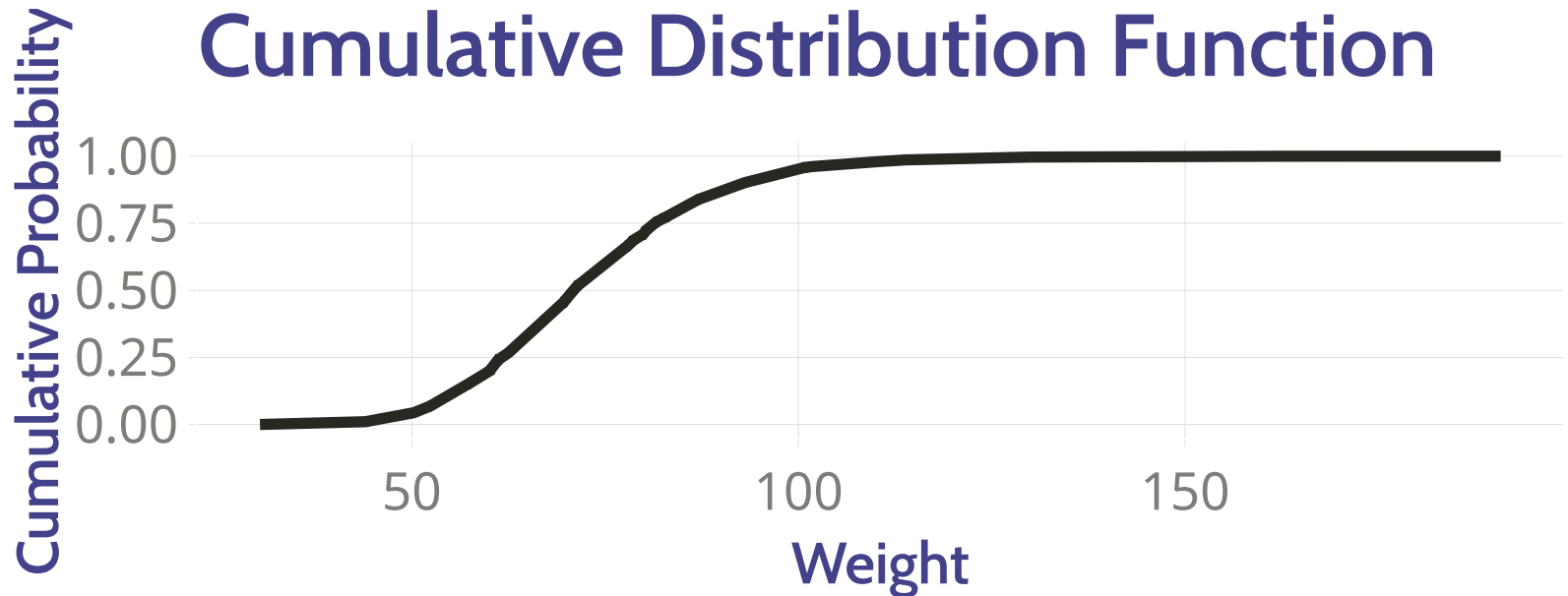
Sorted by weight



Empirical CDF

$$ECDF(x) = \frac{\sum I(w_i \leq x)}{N} = \frac{\text{Number of people with weight} \leq x}{N}$$

- $I(w_i \leq x) = 1$ if person i 's weight is $\leq x$ (*indicator function*); N = sample size.



Empirical CDF (cont.)

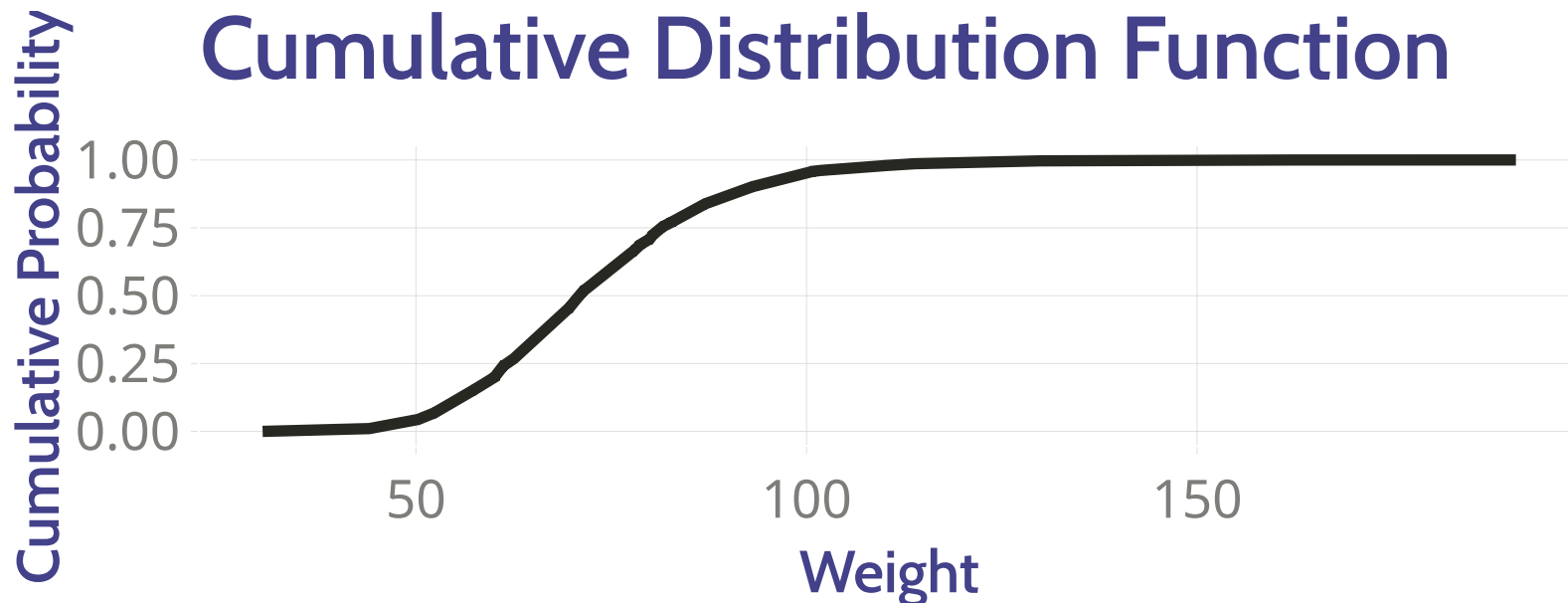
- Share weighing ≤ 50 kg: $P(\text{weight} \leq 50) = ECDF(50)$.
- Share weighing more than 100 kg: $1 - ECDF(100)$.

Empirical CDF: finding percentiles

- Say we want the 25th percentile (1st quartile), call it v_{25}
- By definition $P(X \leq v_{25}) = 25\%$
- So the 1st quartile is the value where the CDF equals 25%
- More generally, percentile Z : $P(X \leq v_Z) = Z\%$

Empirical CDF: finding percentiles (cont.)

Read the percentile off the CDF (interactive):



► [learnr W03_learnr.Rmd](#) → **Exercise 10: Percentiles** — read the 25/50/75/90th off the data.

Appendix — scenario & population vs sample

This week: distributions, fast and hands-on

- This is **review** — you've met distributions before. We move quickly and **do it in R**.
- What a business analyst needs to *read* a distribution: **histograms** and **shape** (modality, skewness).
- We also go a bit deeper on the theory: the **PDF and CDF** sections.
- Behind on the underlying stats? Study those sections; if you're solid, skim them.
- Practice as we go with `W03_learnr.Rmd`.

Summary

- In the last class:
 - We discussed the organization of the course
 - We overviewed forecasting methods
 - We learned about methods of qualitative forecasting
 - *Reference:* Forecasting Methods and Applications, chapter 1
- This set of classes:
 - We will start learning about exploratory analysis preparing the forecast
 - We will learn to translate business problems to statistical problems
 - We will learn about various **data types**
 - We will learn how to **summarize data and observed patterns**
 - *Reference:* Forecasting Methods and Applications, chapter 2.1-2.4

Why do we care?

- To use scientific methods we need to set business problem as a statistical problem
- Define things that we want to find to identify appropriate methods

Specific Scenario

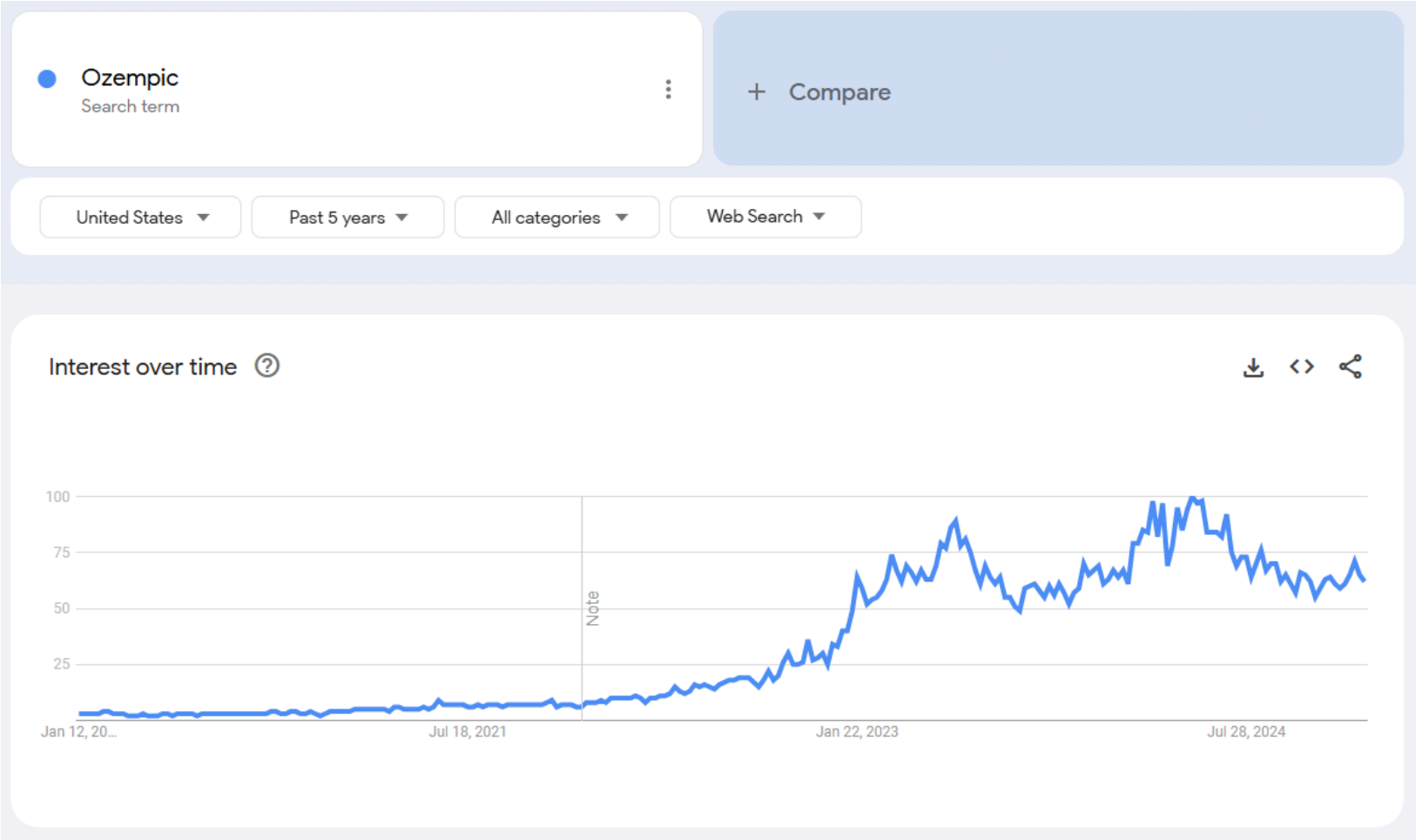
The Opportunity

- Online pharmacies are rapidly growing, offering convenience and accessibility to customers.
 - Example in Mexico: *Choiz*, which provides prescription services and drug subscriptions.

The Trend

- A new generation of highly effective anti-diabetes medications, like *Ozempic*, is gaining popularity for dual benefits: managing diabetes and aiding weight loss.

Search Interest in Ozempic



Your Role

- You have been hired as a consultant for a start-up aiming to launch a subscription service for these medications in Mexico.



The Challenge

- Your boss needs a detailed exploratory market analysis to forecast sales and identify key customer segments

